

Zero auxiliary clipping tables and A196460

Exact counts, all orders asymptotics, and sharp uniform finite sector truncations

Part I: a standalone mathematical research article

Part II: a companion mathematical article

4 October 2026; merged 4 October 2026 (batch 91O)

Abstract

A fixed Boolean clipping table on positive integer inputs is representable by one integer polynomial equation without existential auxiliary variables exactly when every accepted tail cell contains its full coordinate-flat closure. We prove this criterion and a one-positive-witness upper bound, then count the zero-auxiliary tables of side length $n + 1$: their number is $C_n = 1 + a_n$, where a_n is OEIS A196460. Boundary marks identify a_n with an isolated-vertex weighted sum over labeled bipartite graphs. We give explicit polynomial coefficients for every fixed order of its exponentially small corrections, prove a uniform tail estimate, and derive logarithmic and Laurent inverse recurrences. Connected graphs with two named proper colors determine the leading coefficients and the sharp signs of the remainders. A finite smooth proxy and the mean-value theorem justify inversion at sequence values; a separate exact comparison resolves every integer threshold, including the exceptional value $C_0 = 2$. Existing sequence and relational-model interpretations are attributed. All asymptotic orders are fixed, and no convergence, novelty, or theorem-prover verification claim is made.

Keywords. clipping tables; polynomial zero sets; A196460; bipartite graphs; asymptotic inversion; connected components.

Editorial note ([write], 4 October 2026, batch 91O). This report is built from two AI-assisted manuscripts of one external research pipeline, delivered together as manuscripts 21 and 24 of batch 91 of ProveIt’s incoming-reports intake. Manuscript 21 (Part I, the base, whose abstract stands above) proves the zero-versus-one classification of clipping tables, counts the zero-auxiliary tables as $C_n = 1 + a_n$ with $a_n = \text{OEIS A196460}$, and derives every *fixed* order of the forward, logarithmic and inverse expansions and an exact integer threshold rule. Manuscript 24 (Part II) answers Part I’s first further question, “Growing truncation order”, for the exact sector tail: for $n \geq 32$ it bounds every truncation simultaneously, with the optimal constant $9/13$. They are not editions of each other; each Part keeps its own notation, and Section 2 tabulates the letters they use differently. Text added in the write is marked [write]. Sections 1–4 are front matter written for this report; Appendix C records the provenance and every editorial change. The report is unrefereed and not formalized; no statement in it has been checked by a proof assistant.

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1 How the two Parts fit together

[write] Both manuscripts are dated 4 October 2026 and were written in sequence by the same external pipeline. Manuscript 24 embeds the whole release of manuscript 21 unchanged and authenticates it by the SHA-256 of its delivered archive (821a1cfc...); it cites Part I as “the companion article” [7].

- **Part I** (Sections 5–14, Appendix A): the classification of clipping tables (Section 6), the count $C_n = 1 + a_n$ (Section 7), the bipartite-graph reading (Section 8), fixed-order forward asymptotics with an explicit tail (Section 9), logarithmic coefficients and connected two-coloured graphs (Section 10), the sequence-point inverse (Section 11), exact integer thresholds (Section 12), the evidence (Section 13) and four further questions (Section 14).
- **Part II** (Sections 15–22, Appendix B): the same finite sector identity, re-derived (Section 15); complement symmetry and a raising identity; adjacent-ratio bounds on both discrete-Gaussian lattices; the sharp all-order tail bound (Theorem 15.1) and its version for C_n (Theorem 15.2); the exact range $k^2/n \rightarrow 0$ in which a sector coefficient is asymptotic to its leading monomial (Theorem 20.1); and a corrected estimate uniform for $k = o(n^{2/3})$ (Theorem 21.1).

What Part II changes in Part I. Part I’s question “Growing truncation order” (Section 14) is answered for the exact sector tail by Theorems 15.1 and 15.2; it stays open for the logarithmic and inverse expansions of Sections 10 and 11, as Part II says itself (Section 22). Part II also gives a second route to Part I’s sharp remainder (15) (the note after Theorem 9.1). Part I’s other three questions are untouched.

Duplicated material. Part II recalls the count and the graph model without proof (Section 15) and re-derives the finite sector identity (11) as (46), the polynomials (13) as (70) and the leading coefficients (12) as (76). These passages set up Part II’s own notation and are printed in both Parts, each marked in Part II by a [write] note naming the Part I display. No other result is proved in both Parts.

2 Notation across the two Parts

[write] No symbol of either manuscript was renamed. Each Part uses its own manuscript’s letters, defined where they first occur. The polynomials P_k , the coefficients α_k , $\lambda = \log 2$, the expectation $\mathbb{E}$ and the tail $R_{n,M}$ mean the same in both Parts; Part II’s A_n is Part I’s $\mathcal{A}_n$. The letters in Table 1 mean different things in the two Parts; the right-hand column names the tempting false reading.

Table 1: Letters with different meanings in Parts I and II.

Letter	Part I (manuscript 21)	Part II (manuscript 24)	Not to be read as
S	the clipping table $S \subseteq \{1, \dots, K\}^2$	the sector $S(n, k) = P_k(n)2^{-nk}$ and the merged sector $S_C(n, k)$, (45), (50)	Part II’s $S(n, k)$ is a number, not a table
K	the clipping threshold, $K = n + 1$; the tail label	the sharp constant K_n of (48)	$K_n \sim 9/(13n)$ is not $n + 1$

Letter	Part I	Part II	Not to be read as
D	the Laurent inverse coefficients $D_k(s)$, (27)	$D = n - k/2$ in (54)	Part II's D is a half-distance, not an inverse coefficient
B	the majorant $B_{n,k} = \binom{2n}{k} 2^{-nk+k^2/4}$, (16); the binary relation $B(x, y)$ of (10)	the bound B_n of (69); the product $B_{n,r,c}$ of (77)	$B_{n,r,c} \leq 1$ is not the majorant $B_{n,k}$
H	the finite smooth proxy $H_M(x)$, (33)	the tail ratio H_ℓ of (65)	H_ℓ is a rational function of n , not a proxy for $\log_2 a_n$
J	the tail coordinates $J(a)$ of a cell; the leading inverse remainder $J_{n,M}$	the merged tail ratio J_ℓ of (74)	neither J_ℓ nor $J_{n,M}$ is a set of coordinates
q	$q = 2^{-m}$ in (42)	the ratio bound $q_n = 3/(2(n+1))$, (56)	q_n is not exponentially small
U	the accepted set $U_S \subseteq \mathbb{Z}_{\geq 0}^2$; the unary predicates U, U_1 of (10)	$U = k(k-2)/(4n)$, (81)	Part II's U is a number
b	$b_k = k! \alpha_k$, two-coloured graphs (A047863)	$b(h)$, a product of binomials, (57)	$b(h)$ is not b_k (Part II writes $k! \alpha_k$ for b_k)
c	the clipping map c_K ; the number c of column marks; c_k , connected two-coloured graphs (A002027)	a column index c ; the limit $c = \lim k/\sqrt{n}$ in (85)	$e^{-c^2/4}$ involves no c_k
ℓ	a coordinate index $\ell \in \{1, 2\}$	the distance $\ell = 2n - k$ or $2n - j$ from the terminal sector	
h	— (Part I's evidence files use h for $1/\log 2$)	$h = r - k/2$	
d, δ	δ , the formal inverse correction	$d_n = 16n2^{-n/2}$	
E	$\mathbb{E}$, expectation	$\mathbb{E}$, expectation; the relative excess $E_{n,M}, E_{n,M}^C$	$E_{n,M}$ is not an expectation
R	a polynomial $R(A, B)$; R_S ; the set R of marked rows; the part R of the bipartite graph; the tail $R_{n,M}$; the inverse remainder $\mathcal{R}_{n,M}$	the tail $R_{n,M}$, same definition as Part I's	$\mathcal{R}_{n,M}$ is not $R_{n,M}$
m	the threshold index m , and the order index of D_m	a summation index on the half-integer lattice	

Part I's $P_k(x)$ is a polynomial in a real variable (2); Part II's $P_k(n)$ is defined by a sum restricted to $0 \leq r, c \leq n$ (45). They agree at every nonnegative integer n , since $\binom{n}{r} = 0$ for $r > n$; Part I says so after (11). Part II's sector index k runs over $0 \leq k \leq 2n$ for each n , while Part I's expansions fix the order M and let $n \rightarrow \infty$.

3 Relation to other reports and volumes

[write] Neither manuscript names a repository commit, and neither cites any repository text. The following relations were found at the write.

The classification is Report 69's. Part I's Theorem 6.2, its one-positive-witness construction and the remark on dimension at least three in Section 14 are the results of Report 69 of the same

pipeline, *Low-arity finite-clipping Diophantine compilers* (batch-91 manuscript 20): its Section 3 (one witness from disjoint clipped cells), Section 4 (the exact zero-versus-one native classification) and its fixed-dimensional extension. Report 69 is printed as source 38 of Part XIII of the research report `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/signal-machine-collision-certificates` [12], whose text was being written at the same time as this report. Part I’s sources [3, 4] are Report 69’s one-witness note and low-arity audit; they ship there, byte-identical, as `38-low-arity-la-ONE_WITNESS.md` and `38-low-arity-audit-la-AUDIT.md`. Part I says that it reproduces this classification; it is printed here as a restatement with proof, not as a new theorem. What is new in Part I is the count of the closed tables and its asymptotics.

The asymptotic method is the transseries volumes’. Part I’s Sections 9–12 apply, to a different sequence, the method that the companion transseries volume *Combinatorial Transseries and Their Inverses* [13] uses for connected labelled graphs in its Chapter 8, “Connected labelled graphs: every exponential layer” (label `ct:ch:graphs`): exponentially small layers with polynomial coefficients proved by an explicit tail bound (its Theorem 8.1, `t2:thm:graphs`); a formal logarithm of a zero-radius series justified by finite truncations; and an inverse at the sequence values through a finite smooth forward model, with no interpolation of the sequence (its Corollary 8.2, `t2:cor:graph-range`), certified by a residual divided by a slope bound (its Theorem 2.12, `t1:thm:certificate`). Part I’s exact threshold rule (Theorem 12.1) is an instance of the staircase theorem of the canonical volume *Transseries* [14] (Theorem J.43, `p0:thm:staircase`, in its Appendix J, “Apparatus for inverting a rapidly growing function”), realized by the route that Chapter 8 of [13] names for an arbitrary target: bracketing consecutive exact counts. Part I cites neither volume. Its theorems for A196460 are not stated in either volume and are not affected; the method is not claimed as new here, and Part I itself makes no novelty claim. Notes marked [write] at Theorems 9.1, 11.2 and 12.1 give the precise correspondences.

The graph counts are A047863 and A002027. Part I’s $b_k = k!\alpha_k$ and c_k (Section 10) are OEIS A047863 and, for $k \geq 1$, A002027 [8, 9]. Part I does not name the two entries; Part II does (Section 20), and credits the parity-dependent theta asymptotic of $k!\alpha_k$ to Kotesovec. The values printed in Part I’s table agree with both entries (checked 4 October 2026). OEIS data are available under CC BY-SA 4.0; the values printed here are recomputed by the shipped programs, and nothing is copied from the entries beyond these values and the attributions.

The Hilbert’s-tenth programme’s intake review. The research programme maintained in `Computability/HilbertTenthProblem/` reviewed the arrival of both archives in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_new_arithmetic_0d7f51c44.md` (commit `a9ab9a698`) [15]. It read the delivery READMEs in full and, in the articles, only Part I’s abstract and Sections 1 and 10 and Part II’s abstract and Sections 1 and 7 (here Sections 5, 14, 15 and 22). Its findings: both archives “concern the number of finite tables”; Part I separates approximate inversion from the exact integer threshold; Part II’s sharp constants concern finite exact sectors, “not an unrestricted growing-order inverse or simultaneous replacement by leading monomials”; the analytic estimates and the claimed sharpness “were not independently proved”; and neither archive supplies a cheaper paid compiler. Of Report 69 it records that the zero-versus-one classification is “for fixed finite clipping tables with unrestricted degree, not arbitrary c.e. languages”, and that the one-witness product Q_S of (7) is not one residual square: squaring it doubles the degree. All of this agrees with the scope statements of both Parts,

which are printed unchanged.

Elsewhere in the repository. No other report, volume or formal development treats A196460, A047863 or A002027, or counts clipping tables (searched 4 October 2026). No Lean or Rocq declaration corresponds to any statement here.

4 What neither Part claims

[write] Every limitation stated by either manuscript is printed in its place. Collected:

- *Part I.* All asymptotic orders are fixed; constants and the starting index may depend on M ; nothing is uniform for an order growing with n (Remark 9.2). No convergence of $\sum_k P_k(x)2^{-kx}$ at noninteger x is claimed; the ordinary and exponential generating functions of a_n have radius zero and are used formally. No exact floor of a truncated real inverse is claimed; the exact comparison of Theorem 12.1 is the integer answer. The classification concerns fixed finite tables and one polynomial equality; it does not establish global Diophantine minimality or classify composed encodings, and the degree $2|S|$ of (5) need not be minimal. The graph reading is not a claim to originate its combinatorial model. The enclosures of Table 2 are retained evidence, not rerun and not a substitute for proof.
- *Part II.* The threshold $n \geq 32$ is convenient, not minimal; table rows below 32 illustrate only. Exact-sector uniformity gives neither growing-order logarithmic or inverse expansions nor a licence to replace earlier exact coefficients by leading monomials at the next sector’s error scale. No convergence, no exact integer inverse by flooring. The theta asymptotic of $k!\alpha_k$ is Kotesovec’s, recorded in A047863.
- *Both.* No novelty, priority, absence-of-prior-work or exhaustive-search claim; no physical simulation, interpreter, saved schedule or theorem prover was run; the searches recorded in the evidence do not establish originality; preservation records are not WORM storage or trusted timestamps; no external submission. All manuscript, visual and tool reviews in the evidence are model-performed (manuscript 24’s delivery README: “no claim of human review is made”), and the “independent” audits come from the same pipeline. No OEIS submission is drafted or made.

Part I

Zero auxiliary clipping tables: exact counts and all orders asymptotics of A196460

[write] Part I is manuscript 21 of batch 91, the article *Zero auxiliary clipping tables: Exact counts and all orders asymptotics of A196460*, author line “A standalone mathematical research article”, dated 4 October 2026 (16 pages as delivered). Its abstract opens this report; its text follows unchanged except for the label prefix **cta:** and the [write] notes.

5 The count and the questions it answers

Let $K \geq 1$ be a fixed integer and let $S \subseteq \{1, \dots, K\}^2$ be a Boolean table, with membership interpreted as acceptance. Define

$$c_K(x) = \min(x, K), \quad U_S = \{(A, B) \in \mathbb{Z}_{>0}^2 : (c_K(A), c_K(B)) \in S\}.$$

The label K is a *tail*: it represents every integer at least K . A label $i < K$ represents only the integer i . We ask for the smallest number of positive existential auxiliary variables in a single integer-coefficient polynomial equation representing U_S . The table is fixed coefficient data. Its size is not an input; its coefficients and polynomial degree may depend on the entire table. No degree bound is imposed.

Put $n = K - 1$, and let C_n be the number of tables whose minimum is zero. The exact answer is

$$C_n = 1 + a_n, \quad a_n = \sum_{r,c=0}^n \binom{n}{r} \binom{n}{c} 2^{(n-r)(n-c)} = \sum_{j=0}^n \binom{n}{j} (1 + 2^j)^n. \quad (1)$$

The last sum is the defining term formula in OEIS A196460 [1]. We prove the classification and count in Sections 6 and 7; every table not counted by C_n has minimum exactly one.

The dominant count is 2^{n^2} , but the corrections have a useful explicit structure. For $k \geq 0$ define the polynomial

$$P_k(x) = \sum_{r=0}^k \binom{x}{r} \binom{x}{k-r} 2^{r(k-r)}, \quad \binom{x}{r} = \frac{x(x-1) \cdots (x-r+1)}{r!}, \quad \binom{x}{0} = 1. \quad (2)$$

For each *fixed* $M \geq 0$, as $n \rightarrow \infty$ through integers,

$$\frac{a_n}{2^{n^2}} = \sum_{k=0}^M P_k(n) 2^{-kn} + O_M(n^{M+1} 2^{-(M+1)n}). \quad (3)$$

The error is nonnegative and its leading equivalent is explicit. The resulting zero-auxiliary fraction among all $2^{(n+1)^2}$ tables is asymptotic to $\frac{1}{2} 4^{-n}$.

For inversion set $\lambda = \log 2$, $s_n = \sqrt{\log_2 a_n}$ and $t_n = 2^{-s_n}$. There are explicit Laurent polynomials D_k , beginning with $D_1 = -1/\lambda$, such that for every fixed $M \geq 1$,

$$n = s_n + \sum_{k=1}^M D_k(s_n) t_n^k + O_M(s_n^M t_n^{M+1}). \quad (4)$$

Sections 10–11 supply the recurrences and all analytic error arguments. Section 12 explains why a real asymptotic inverse must be distinguished from an exact integer inverse.

Relation to earlier work. The OEIS entry credits Paul D. Hanna, 2 October 2011, for the sequence and Václav Kotesovec, 25 June 2013, for $a_n \sim 2^{n^2}$ [1]. Svatoš, Jung, Tóth, Wang and Kuželka list A196460 with a first-order relational sentence in Appendix A.1, Table 3, printed page 32 of their 2023 paper [2]. Section 8 proves how its active-vertex interpretation gives the weighted bipartite graph count. The clipping-table interpretation uses the separately established coordinate-flat criterion, reproduced here with its one-witness construction [3, 4]. The present exposition and asymptotic proofs organize accepted mathematical source and independent audit material [5, 6]. No priority or absence-of-prior-work assertion is intended.

Attributions added in this report. [write] The classification of Section 6 is the theorem of Report 69 of the same pipeline, printed as source 38 of Part XIII of the report signal-machine-collision-certificates [12]; the sources [3, 4] are that Report’s one-witness note and low-arity audit. The method of Sections 9–12 is that of Chapter 8 of the companion transseries volume [13] and of the staircase theorem of the canonical volume [14]; the integers b_k and c_k of Section 10 are OEIS A047863 and A002027 [8, 9]. Section 3 gives the details; none of this changes a statement of this Part.

6 A self contained zero versus one classification

For a cell $a = (a_1, a_2)$ write

$$J(a) = \{\ell \in \{1, 2\} : a_\ell = K\}.$$

Its coordinate-flat closure is obtained by replacing any or all coordinates in $J(a)$ by arbitrary labels from $\{1, \dots, K\}$, while leaving all other coordinates fixed. Call S *closed* if it contains this closure for every $a \in S$. Concretely, an accepted (i, K) with $i < K$ forces the whole row i , an accepted (K, j) with $j < K$ forces the whole column j , and an accepted (K, K) forces the full table. Finite coordinates are not replaced: this is not the usual order-ideal condition on all coordinates.

Lemma 6.1 (An infinite grid determines a polynomial). *If a real polynomial in d variables vanishes on $E_1 \times \dots \times E_d$, with every $E_j \subseteq \mathbb{R}$ infinite, then it is the zero polynomial.*

Proof. For $d = 1$, a nonzero polynomial has only finitely many roots. Induct on d . Write the polynomial as $\sum_{q=0}^D f_q(X_1, \dots, X_{d-1})X_d^q$. Fixing $(X_1, \dots, X_{d-1})$ in the product of the first $d - 1$ sets gives a univariate polynomial zero on E_d , so every coefficient f_q is zero at that fixed tuple. Thus each f_q vanishes on the full product of the first $d - 1$ sets, and the induction hypothesis makes it identically zero. $\square$

Theorem 6.2 (Exact auxiliary minimum). *For a fixed clipping table S , the following conditions are equivalent:*

- (i) *There is an $R \in \mathbb{Z}[A, B]$ such that $R(A, B) = 0$ on positive integer inputs exactly when $(A, B) \in U_S$.*
- (ii) *The table S is closed under coordinate-flat replacement.*

Every table has a representation by one polynomial equality with one positive existential auxiliary variable. Consequently its exact minimum is zero if it is closed, and one otherwise.

Necessity and the zero-auxiliary construction. Suppose R represents U_S and $a \in S$. Fix each coordinate outside $J(a)$ at its label a_ℓ . The remaining variables range independently over $\{K, K + 1, \dots\}$, and the restricted polynomial vanishes at every point of that infinite grid. Lemma 6.1 says that the restriction is identically zero. It therefore also vanishes when former tail coordinates are replaced by arbitrary positive integers below K . Exact representation forces all their clipped cells to remain accepted. This proves closure, and works even for real coefficients without any nonnegativity assumption on R .

Conversely, put $X_1 = A$, $X_2 = B$ and define

$$Z_a(X) = \sum_{\ell \notin J(a)} (X_\ell - a_\ell)^2, \quad R_S(X) = \prod_{a \in S} Z_a(X). \quad (5)$$

An empty sum is 0 and an empty product is 1. Thus the empty table has $R_S = 1$. If the all-tail cell is accepted, closure makes the table full and its factor is 0, giving $R_S = 0$ as required. In the remaining cases Z_a vanishes precisely on the coordinate flat with the finite coordinates fixed. At any positive point of that flat, the clipped cell is one of the allowed replacements of a , so closure prevents unintended acceptance. Conversely, an accepted input makes the factor of its own clipped cell vanish. A product is zero exactly when one of its factors is zero, proving the desired equivalence. $\square$

The one-positive-witness upper bound. For $K \geq 2$ define

$$p_K(x) = \prod_{h=1}^{K-1} (x - h)^2. \quad (6)$$

For positive integers x this is zero exactly when $x < K$, and strictly positive exactly when $x \geq K$. For each cell set

$$f_a(X, w) = \sum_{\ell \notin J(a)} (X_\ell - a_\ell)^2 + \left(w - \prod_{\ell \in J(a)} p_K(X_\ell) \right)^2, \quad Q_S(X, w) = \prod_{a \in S} f_a(X, w), \quad (7)$$

where the inner empty product is 1. Every factor is a sum of squares. For positive integers X_1, X_2, w , its vanishing forces all finite coordinates to have their prescribed labels. It also forces the product of the tail polynomials to equal $w > 0$. Each factor $p_K(X_\ell)$ is nonnegative, so all tail coordinates are at least K . Conversely, on that clipped cell, choosing w to be the displayed product gives a positive integer zero. Thus $f_a = 0$ holds exactly on cell a with this one witness value.

The product Q_S is zero if and only if an accepted cell factor is zero. Clipped cells partition $\mathbb{Z}_{>0}^2$, so the positive witness is unique for every accepted input. For $K = 1$, the only possibilities are the full or empty table; use $(w - 1)^2$ or 1, respectively. Both also have zero-auxiliary representations 0 and 1. This proves the upper bound and, together with necessity, the exact minimum asserted. $\square$

Strict positivity of the witness is essential in (7): if $w = 0$ were allowed, a below-threshold coordinate could make its tail polynomial zero and impersonate a tail. These are fixed finite-table statements. They neither establish global Diophantine minimality nor classify the auxiliary count of a larger composed encoding.

[write] Theorem 6.2 and both proofs above restate Report 69's Section 4 (the zero-versus-one native classification) and Section 3 (one witness from disjoint clipped cells), printed as source 38 of [12]; see Section 3. The Hilbert's-tenth programme's intake review [15] adds two scope remarks, both consistent with the paragraph above: the classification is for fixed finite clipping tables with unrestricted degree, not for arbitrary recursively enumerable sets; and Q_S in (7) is a product of nonnegative cell polynomials, not one residual square, so writing it as a square doubles its degree.

[write] (Batch 91, reciprocal note, 4 October 2026.) Part XIII of [12] is now written (commit 61c9e3eb2). There Report 69's Section 4 is Theorem `smc:la:thm:classification`, in Section `smc:la:sec:classification`, which also holds its fixed-dimensional extension; its Section 3 is Section `smc:la:sec:one`; and its coefficient, storage and circuit costs are Section `smc:la:sec:costs`. A dated note at the end of Section `smc:la:sec:classification` points back to this report's count.

7 Counting tables by their boundary marks

Assume the corner (K, K) is rejected. Let R be the set of the n finite row labels i for which (i, K) is accepted, and T the set of the n finite column labels j for which (K, j) is accepted. These marks

are read from the boundary, regardless of whether unmarked interior rows or columns happen to be full. If $|R| = r$ and $|T| = c$, closure forces every interior entry in a marked row or column. Precisely $(n - r)(n - c)$ interior entries remain independent Boolean choices. The boundary and rejected corner are already fixed.

	1	2	3	4	K	
1	*	*	1	*	0	
2	*	*	1	*	0	
3	*	*	1	*	0	
4	1	1	1	1	1	Accepted boundary marks force the highlighted row and column
K	0	0	1	0	0	column mark

$n = 4, r = c = 1$
Nine free interior entries
give 2^9 tables for these marks

Figure 1: Boundary data, not accidental full interior rows, define the marks. A $*$ is an arbitrary bit. The corner is rejected; accepting it gives the single additional full table.

There are $\binom{n}{r}\binom{n}{c}$ choices of marks. Summing over r, c and adding the sole table with accepted corner proves the first identity in (1). To obtain the second, put $j = n - r$ and use

$$\sum_{c=0}^n \binom{n}{c} (2^j)^{n-c} = (1 + 2^j)^n.$$

This also proves the formula at $n = 0$: the double sum is 1, and $C_0 = 2$. The initial counts are

n	a_n	C_n
0	1	2
1	5	6
2	47	48
3	1 193	1 194
4	113 855	113 856
5	46 857 665	46 857 666
6	83 540 629 607	83 540 629 608

Different boundary patterns on an all-one interior are different tables. They are correctly counted separately; only the full table with accepted corner is supplied by the additional 1.

8 Bipartite graphs and the existing relational model

Take two distinguished labeled parts L and R , each of size n . A graph is any subset of $L \times R$, so there are n^2 independently available edges. Write $i(G)$ for the total number of isolated vertices.

Proposition 8.1. *The sequence has the weighted graph interpretation*

$$a_n = \sum_G 2^{i(G)}. \quad (8)$$

For a uniform random graph in this model,

$$\frac{a_n}{2^{n^2}} = \mathbb{E}[2^{i(G)}], \quad \mathbb{E}\binom{i(G)}{k} = P_k(n)2^{-nk}. \quad (9)$$

Proof. Count pairs (G, U) where U is a subset of the isolated vertices. Choose r marked left vertices and c marked right vertices. Their incident edges must be absent and the other $(n-r)(n-c)$ edges are free, giving exactly the double sum in (1). For each fixed graph there are $2^{i(G)}$ mark sets.

Directly, put an edge at each *rejected* interior table entry. An accepted boundary mark can only occur at an isolated vertex, and any subset of isolated vertices can be marked. This is a bijection with the closed tables having rejected corner.

For the second expectation, a specified collection of r left and $k-r$ right vertices is isolated precisely when $nk - r(k-r)$ distinct edges are absent. This event has probability $2^{-nk+r(k-r)}$. Sum over all choices of the collection, using (2). The first expectation follows by dividing (8) by 2^{n^2} . $\square$

Svatoš et al. explicitly associate A196460 with the sentence

$$(\forall x \forall y [U(x) \vee B(x, y)]) \wedge (\forall x \forall y [B(x, y) \vee U_1(y)]) \quad (10)$$

in Appendix A.1, Table 3, printed page 32 of version 1 [2]. The independent unary predicates U, U_1 choose active subsets of the left and right copies of the domain. The relation B must be true except possibly at a pair whose two copies are active. Interpret a false entry of B as an edge. For any fixed graph, its nonisolated vertices must be active, while each isolated vertex can be active or inactive, yielding the weight $2^{i(G)}$ again. Although the relational structure uses one domain, the graph uses two distinguished copies: $B(x, x)$ corresponds to an edge between copies, not a graph loop. All n^2 edge choices remain available. This graph reading is a direct interpretation of the existing sentence, not a claim to originate its combinatorial model.

9 All fixed order forward asymptotics

Dividing (1) by 2^{n^2} and grouping by $k = r + c$ gives the exact finite identity

$$\mathcal{A}_n := \frac{a_n}{2^{n^2}} = \sum_{k=0}^{2n} P_k(n)2^{-kn}. \quad (11)$$

At a nonnegative integer n , $\binom{n}{r} = 0$ for $r > n$, so the polynomial formula (2) is valid even when $k > n$, and $P_k(n) = 0$ for $k > 2n$. The same equality follows by averaging the finite identity $2^{i(G)} = \sum_k \binom{i(G)}{k}$ in (9).

Every P_k has exact degree k and positive leading coefficient

$$\alpha_k = \sum_{r=0}^k \frac{2^{r(k-r)}}{r!(k-r)!}. \quad (12)$$

The first five polynomials are

$$\begin{aligned} P_0(x) &= 1, & P_1(x) &= 2x, \\ P_2(x) &= 3x^2 - x, & P_3(x) &= \frac{13}{3}x^3 - 5x^2 + \frac{2}{3}x, \\ P_4(x) &= \frac{27}{4}x^4 - \frac{33}{2}x^3 + \frac{41}{4}x^2 - \frac{1}{2}x. \end{aligned} \quad (13)$$

An exact finite regrouping alone is insufficient for an asymptotic expansion: its number of terms grows with n . We next control that tail.

Theorem 9.1 (Uniform tail for every fixed order). *For a fixed $M \geq 0$ let $R_{n,M} = \mathcal{A}_n - \sum_{k=0}^M P_k(n)2^{-kn}$. If $n \geq M+1$ and $\rho_n := 2n2^{-n/2+1/4} < 1$, then*

$$0 \leq R_{n,M} \leq \frac{(2n)^{M+1}2^{-(M+1)n+(M+1)^2/4}}{(M+1)!(1-\rho_n)} + 2^{2n-3n^2/4}. \quad (14)$$

In particular (3) holds. Its remainder is sharp:

$$R_{n,M} \sim \alpha_{M+1}n^{M+1}2^{-(M+1)n}. \quad (15)$$

Proof. Since $r(k-r) \leq k^2/4$, Vandermonde's identity gives, for $0 \leq k \leq 2n$,

$$0 \leq P_k(n)2^{-kn} \leq B_{n,k} := \binom{2n}{k} 2^{-nk+k^2/4}. \quad (16)$$

For $0 \leq k < n$,

$$\frac{B_{n,k+1}}{B_{n,k}} = \frac{2n-k}{k+1} 2^{-n+k/2+1/4} \leq 2n2^{-n/2+1/4} = \rho_n.$$

Consequently the sum over $M+1 \leq k \leq n$ is at most $B_{n,M+1}/(1-\rho_n)$. The inequality $\binom{2n}{M+1} \leq (2n)^{M+1}/(M+1)!$ yields the first term of (14).

On $[n, 2n]$ the function $-nk + k^2/4$ is decreasing, with value $-3n^2/4$ at $k = n$. Hence

$$\sum_{k=n+1}^{2n} B_{n,k} \leq 2^{-3n^2/4} \sum_{k=n+1}^{2n} \binom{2n}{k} \leq 2^{2n-3n^2/4}.$$

This proves the explicit bound. Since $\rho_n \rightarrow 0$, the first term is $O_M(n^{M+1}2^{-(M+1)n})$. The second is little-oh of that scale: the logarithm of their ratio has leading term $-3(\log 2)n^2/4$. This proves (3). For a concrete uniform range, $n \geq \max(M+1, 16)$ permits replacing $1/(1-\rho_n)$ by 2: $\rho_{16} < 1/2$ and ρ_n decreases thereafter.

Finally apply the already proved expansion with order $M+1$ and subtract the truncation at M . Its first omitted term is $P_{M+1}(n)2^{-(M+1)n} \sim \alpha_{M+1}n^{M+1}2^{-(M+1)n}$. The next error divided by this scale is $O_M(n2^{-n}) \rightarrow 0$. This proves (15). $\square$

[write] *Relation to the transseries volume.* Theorem 9.1 is the A196460 analogue of Theorem 8.1 of [13] (`t2:thm:graphs`), which proves every fixed exponential layer of $c_n/2^{\binom{n}{2}}$ for connected labelled graphs with explicit polynomial coefficients by the same device: a majorant whose successive ratios are eventually below a constant less than one, plus a separate bound on the quadratically small far range. That theorem is not cited by manuscript 21.

[write] *Second route to (15) (Part II).* Part II's $R_{n,M}$ is the same tail. For $n \geq 32$, Theorem 15.1 gives $R_{n,M} = P_{M+1}(n)2^{-(M+1)n}(1 + E_{n,M})$ with $0 \leq E_{n,M} \leq K_n = O(1/n)$, uniformly in $0 \leq M < 2n$; since P_{M+1} has degree $M+1$ and leading coefficient α_{M+1} , this gives (15) again for each fixed M , and Theorem 20.1 extends the leading equivalent to every order $M = M(n)$ with $M^2/n \rightarrow 0$.

Remark 9.2 (Fixed order does not mean convergence). The constants and the sufficiently large starting index may depend on M . No estimate here is uniform for a truncation order growing with n . For each integer n , (11) is finite; no claim is made that $\sum_{k \geq 0} P_k(x)2^{-kx}$ converges for noninteger x . Formal power series used below require no such convergence.

9.1 The rarity of zero auxiliary arity

The exact fraction of zero-auxiliary tables is

$$F_n = \frac{C_n}{2^{(n+1)^2}} = 2^{-2n-1} (\mathcal{A}_n + 2^{-n^2}). \quad (17)$$

The additional full table contributes the isolated 2^{-n^2} inside the parentheses. It is smaller than every fixed exponential sector as $n \rightarrow \infty$. The first terms are therefore

$$F_n = 2^{-2n-1} \left[1 + 2n2^{-n} + (3n^2 - n)2^{-2n} + O(n^3 2^{-3n}) \right] \sim \frac{1}{2} 4^{-n} = 2^{1-2K}. \quad (18)$$

The complementary fraction $1 - F_n$ has exact minimum one positive auxiliary by Theorem 6.2. At $n = 0$ the exact formula gives $F_0 = 1$, as both one-cell tables have minimum zero.

10 Logarithmic coefficients and connected colored graphs

In the formal ring $\mathbb{Q}[x][[z]]$ define L_k by

$$\log \left(1 + \sum_{k \geq 1} P_k(x) z^k \right) = \sum_{k \geq 1} L_k(x) z^k. \quad (19)$$

Differentiating with respect to z and multiplying by the series inside the logarithm gives the explicit recurrence

$$L_k = P_k - \frac{1}{k} \sum_{j=1}^{k-1} j L_j P_{k-j} \quad (k \geq 1). \quad (20)$$

Induction shows $\deg L_k \leq k$. The first four are

$$\begin{aligned} L_1(x) &= 2x, & L_2(x) &= x^2 - x, \\ L_3(x) &= x^3 - 3x^2 + \frac{2}{3}x, \\ L_4(x) &= \frac{19}{12}x^4 - \frac{15}{2}x^3 + \frac{101}{12}x^2 - \frac{1}{2}x. \end{aligned} \quad (21)$$

Proposition 10.1 (Actual logarithmic expansion). *For every fixed $M \geq 1$,*

$$\log_2 a_n = n^2 + \frac{1}{\lambda} \sum_{k=1}^M L_k(n) 2^{-kn} + O_M(n^{M+1} 2^{-(M+1)n}), \quad \lambda = \log 2. \quad (22)$$

Proof. Theorem 9.1 at order zero gives $u_n := \mathcal{A}_n - 1 = O(n 2^{-n}) \rightarrow 0$. For fixed M it also gives $u_n = \sum_{k=1}^M P_k(n) 2^{-kn} + O_M(n^{M+1} 2^{-(M+1)n})$. The derivative of $\log(1 + u)$ is bounded near $u = 0$, so substituting this truncation into the logarithm changes it by the same error order. Its Taylor polynomial through power M has remainder $O(u_n^{M+1}) = O_M(n^{M+1} 2^{-(M+1)n})$.

Expand that finite Taylor polynomial. A term with total exponential index r has polynomial degree at most r . All retained terms of index $r \leq M$ combine to $L_r(n) 2^{-rn}$ by (19). There are only finitely many other terms, all with $r \geq M + 1$; they are bounded by $O_M(n^{M+1} 2^{-(M+1)n})$ because $n 2^{-n} \rightarrow 0$. Finally divide by λ and add n^2 . $\square$

To understand the top coefficients, let $b_k = k! \alpha_k$, with $b_0 = 1$. This counts simple graphs on k labeled vertices equipped with a proper coloring by two *named* colors: choose the r labels of the first color and any of the $r(k-r)$ possible cross-color edges. These graphs have a variable color split; they are not the balanced n -by- n ambient graphs counted in (8).

Let c_k count the connected such colored graphs, where a singleton has two color choices. A colored graph decomposes uniquely into its connected colored components. Summing over partitions of its label set gives

$$\sum_{k \geq 0} b_k \frac{z^k}{k!} = \exp \left(\sum_{k \geq 1} c_k \frac{z^k}{k!} \right) \quad (23)$$

formally. More explicitly, the coefficient at $z^k/k!$ on the right sums $\prod_{B \in \pi} c_{|B|}$ over every set partition π of the k labels, which is exactly the component decomposition. Thus no analytic convergence or random-graph connectivity theorem enters this identity.

Choosing the connected component containing label 1 gives a constructive recurrence:

$$c_k = b_k - \sum_{j=1}^{k-1} \binom{k-1}{j-1} c_j b_{k-j}. \quad (24)$$

Since $\deg P_k = k$, taking top-degree coefficients in (19) amounts to taking the formal logarithm of $\sum \alpha_k z^k$. Equation (23) therefore proves

$$[x^k] L_k(x) = \frac{c_k}{k!} > 0. \quad (25)$$

Positivity follows directly: the singleton works for $k = 1$, and a properly colored star supplies a connected example for every $k \geq 2$. For reference, the exact values retained by the independent audit are

k	b_k	c_k
1	2	2
2	6	2
3	26	6
4	162	38
5	1 442	390
6	18 306	6 062
7	330 626	134 526
8	8 488 962	4 172 198

[write] These are OEIS A047863 (b_k , labelled graphs with two named proper colours) and A002027 (c_k for $k \geq 1$; the entry's value at $k = 0$ is 1, a separate convention) [8, 9]; manuscript 21 does not name them, manuscript 24 does (Section 20). The tabulated values agree with both entries, and (23) is the relation between their exponential generating functions recorded in A002027.

Applying Proposition 10.1 at one additional order also gives the sharp logarithmic remainder

$$\log_2 a_n - n^2 - \frac{1}{\lambda} \sum_{k=1}^M L_k(n) 2^{-kn} \sim \frac{c_{M+1}}{\lambda(M+1)!} n^{M+1} 2^{-(M+1)n}. \quad (26)$$

In particular its sign is eventually positive for each fixed M .

11 Formal inverse coefficients with an actual error proof

We first define a formal series; we then justify it on the actual sequence values without introducing an interpolation of the sequence. For coefficient extraction treat s and t as independent and work in

$$\mathbb{Q}[\lambda, \lambda^{-1}, s, s^{-1}][[t]].$$

Here λ is a formal symbol in the algebra; subsequently it is evaluated at $\log 2 > 0$. Put $\delta_0 = 0$ and, recursively,

$$\begin{aligned} \delta_{m-1}(s, t) &= \sum_{j=1}^{m-1} D_j(s) t^j, \\ D_m(s) &= -\frac{1}{2s} [t^m] \left\{ \delta_{m-1}(s, t)^2 \right. \\ &\quad \left. + \frac{1}{\lambda} \sum_{k=1}^m L_k(s + \delta_{m-1}(s, t)) t^k e^{-k\lambda\delta_{m-1}(s, t)} \right\}. \end{aligned} \quad (27)$$

Only finitely many Taylor coefficients contribute at each step. Equivalently, $\delta = \sum_{j \geq 1} D_j(s) t^j$ uniquely solves the formal equation

$$2s\delta + \delta^2 + \frac{1}{\lambda} \sum_{k \geq 1} L_k(s + \delta) t^k e^{-k\lambda\delta} = 0. \quad (28)$$

At order t^m , the new D_m occurs only in $2s\delta$. The square has no constant term, and each summand already carries a positive power of t . The coefficient of D_m is therefore $2s$, invertible in this Laurent ring, proving both the recurrence and uniqueness.

The first three coefficients are

$$\begin{aligned} D_1(s) &= -\frac{1}{\lambda}, \\ D_2(s) &= -\frac{s+1}{2\lambda} + \frac{1}{2\lambda^2 s}, \\ D_3(s) &= -\frac{s^2}{2\lambda} - \frac{1}{3\lambda} + \frac{1}{\lambda^2} + \frac{1}{\lambda^2 s}. \end{aligned} \quad (29)$$

The negative powers are harmless in the asymptotic region $s \rightarrow \infty$; these formulas are not meant for evaluation at $s = 0$.

Lemma 11.1 (Degree bound and sign of the leading coefficient). *The largest exponent of s in D_m is $m - 1$, and*

$$D_m(s) \sim -\frac{c_m}{2\lambda m!} s^{m-1}. \quad (30)$$

In particular $D_m(s) = O_m(s^{m-1})$ for $s \geq 1$ as $s \rightarrow \infty$.

Proof. Induct on m . Assume every earlier D_j has largest s exponent at most $j - 1$. A product of p earlier inverse coefficients with total t -degree $m - k$ has largest s exponent at most $m - k - p$. Derivatives of the polynomial L_k can only lower its degree from at most k ; the exponential introduces powers of λ but no new powers of s . Thus every coefficient at t^m in the sum of (27) has exponent at most m before division by s . If any earlier D_j occurs, then $p \geq 1$ and the bound is strictly less than m . The term δ_{m-1}^2 has exponent at most $m - 2$. The only possible contribution at exponent m is therefore $\lambda^{-1} L_m(s)$, whose coefficient is $c_m/(\lambda m!)$ by (25). Division by $-2s$ proves both the exact top degree and (30). This also covers $m = 1$. $\square$

Theorem 11.2 (Sequence-point inversion and its sharp sign). *For $s_n = \sqrt{\log_2 a_n}$, $t_n = 2^{-s_n}$ and each fixed $M \geq 1$,*

$$n - \left(s_n + \sum_{k=1}^M D_k(s_n) t_n^k \right) = O_M(s_n^M t_n^{M+1}). \quad (31)$$

More sharply,

$$n - \left(s_n + \sum_{k=1}^M D_k(s_n) t_n^k \right) \sim -\frac{c_{M+1}}{2\lambda(M+1)!} s_n^M t_n^{M+1}. \quad (32)$$

Every fixed truncation consequently overestimates the integer n for all sufficiently large n .

Proof. Fix M and introduce the *finite* smooth function

$$H_M(x) = x^2 + \frac{1}{\lambda} \sum_{k=1}^M L_k(x) 2^{-kx}. \quad (33)$$

This is an explicit approximation, not an interpolation of a_n . Proposition 10.1 gives at the actual integer n

$$H_M(n) - s_n^2 = O_M(n^{M+1} 2^{-(M+1)n}). \quad (34)$$

It also gives $s_n^2 - n^2 = O(n 2^{-n})$. Division by $s_n + n$, which is at least $2n$ because $a_n \geq 2^{n^2}$, yields

$$s_n - n = O(2^{-n}), \quad n \sim s_n, \quad 2^{-n}/t_n \rightarrow 1. \quad (35)$$

Define $\hat{n}_M = s_n + \delta_M(s_n, t_n)$, where $\delta_M(s, t) = \sum_{j=1}^M D_j(s) t^j$. Lemma 11.1 and $st \rightarrow 0$ imply $\delta_M(s, t) = O_M(t)$ along $t = 2^{-s}$. Hence both n and $\hat{n}_M$ lie in $[s_n - 1, s_n + 1]$ for all sufficiently large n .

We detail the residual bound at $\hat{n}_M$, since purely formal cancellation is not an analytic error estimate. Substitute $s + \delta_M$ into (33), set $t = 2^{-s}$, and subtract s^2 :

$$2s\delta_M + \delta_M^2 + \frac{1}{\lambda} \sum_{k=1}^M L_k(s + \delta_M) t^k e^{-k\lambda\delta_M}. \quad (36)$$

Expand the finite polynomials exactly. In the term with prefactor t^k , expand the exponential through power $M - k$ of δ_M . The exponential remainder is $O_M(\delta_M^{M-k+1})$, uniformly for small δ_M . As $L_k(s + \delta_M) = O_M(s^k)$ and $\delta_M = O_M(t)$, its contribution is $O_M(s^k t^{M+1})$, hence $O_M(s^{M+1} t^{M+1})$.

The remaining expression is a finite polynomial in t with Laurent coefficients in s . All coefficients through order M vanish by (27). Every remaining coefficient of t -degree r has largest s exponent at most r : for $2s\delta_M$ this follows from Lemma 11.1; for δ_M^2 the stronger bound $r - 2$ holds; and for each expanded logarithmic summand the same product-degree argument as in that lemma gives at most r . Thus each of its finitely many terms with $r \geq M + 1$ is $O_M(s^r t^r) = O_M(s^{M+1} t^{M+1})$, since $s \geq 1$ and $st \rightarrow 0$. Negative powers only improve this bound. We have proved

$$H_M(\hat{n}_M) - s_n^2 = O_M(s_n^{M+1} t_n^{M+1}). \quad (37)$$

Differentiating only the finite function (33), uniformly on $[s_n - 1, s_n + 1]$, gives

$$H'_M(x) = 2x + O_M(x^M 2^{-x}) \geq s_n$$

for large n . Subtract (34) from (37), use (35) to compare the error scales, and apply the mean-value theorem between n and $\hat{n}_M$. Dividing by the derivative lower bound s_n proves (31).

Finally use the proved bound at order $M + 1$. The difference at order M is $D_{M+1}(s_n)t_n^{M+1} + O_M(s_n^{M+1}t_n^{M+2})$. The ratio of the last error to $s_n^M t_n^{M+1}$ tends to zero because $s_n t_n \rightarrow 0$. Lemma 11.1 proves (32) and its negative sign. $\square$

[write] *Relation to the transseries volume.* The proof is the method of Corollary 8.2 of [13] (**t2:cor:graph-range**): a finite smooth forward model, the residual of the formal inverse at the sequence value, and division by a lower bound for the slope, which is the local inverse certificate of its Theorem 2.12 (**t1:thm:certificate**); in both, the quadratic core needs no Lambert function and no interpolation of the sequence is used. Here the core is x^2 in the coordinate $s = \sqrt{\log_2 a_n}$, there $\frac{\log 2}{2}x(x-1)$. The sharp signed remainder (32) has no counterpart there.

11.1 Replacing the sequence by the table count

Since $C_n = a_n + 1$ and $a_n \geq 2^{n^2}$,

$$\log_2 C_n - \log_2 a_n = \log_2(1 + a_n^{-1}) = O(2^{-n^2}). \quad (38)$$

For $n \rightarrow \infty$, the square-root difference is $O(2^{-n^2}/n)$. Both are smaller than every fixed error sector above. Thus all fixed-order forward, logarithmic and inverse expansions, including their sharp remainder equivalents, hold with C_n in place of a_n and $s_n = \sqrt{\log_2 C_n}$. This is an asymptotic statement; it does not erase the different exact endpoint at $n = 0$.

12 Exact integer thresholds and the floor boundary

The double sum immediately gives, for every $n \geq 0$,

$$2^{n^2} \leq a_n \leq 2^{n^2+2n} < 2^{(n+1)^2}. \quad (39)$$

The lower bound is the term $r = c = 0$. For the upper bound replace each $2^{(n-r)(n-c)}$ by 2^{n^2} and sum the binomial weights to obtain 4^n . Consequently

$$\left\lfloor \sqrt{\log_2 a_n} \right\rfloor = n \quad \text{for every } n \geq 0, \quad (40)$$

including $a_0 = 1$. Also a_n is strictly increasing: each old $j \leq n$ term of the single sum for a_{n+1} is at least twice the corresponding term for a_n , because $\binom{n+1}{j} \geq \binom{n}{j}$ and $1 + 2^j \geq 2$; there is an additional positive term at $j = n + 1$.

Theorem 12.1 (One exact comparison suffices). *For real $y \geq 1$ put $m = \lfloor \sqrt{\log_2 y} \rfloor$. Then*

$$N_a(y) := \max\{n \geq 0 : a_n \leq y\} = \begin{cases} m - 1, & y < a_m, \\ m, & y \geq a_m. \end{cases} \quad (41)$$

At $m = 0$ only the second branch occurs. Below $y = 1$ the defining set is empty. For C_n , the same two-branch rule with C_m in place of a_m holds for all $y \geq 2$; below 2 its defining set is empty.

Proof. If $n < m$, then (39) gives $a_n < 2^{(n+1)^2} \leq 2^{m^2} \leq y$. If $n > m$, then $a_n \geq 2^{n^2} \geq 2^{(m+1)^2} > y$. Only index m remains undecided, and its exact comparison yields (41). When $m = 0$, $y \geq 1 = a_0$, so the first branch cannot arise.

For $n \geq 1$, adding one to (39) still gives $2^{n^2} < C_n < 2^{(n+1)^2}$, since $2^{n^2+2n} + 1 < 2^{n^2+2n+1}$. But $C_0 = 2$ equals 2^{1^2} . If $y \geq 2$, then $m \geq 1$. We have $C_{m-1} \leq 2^{m^2} \leq y$, with equality in the first inequality possible at $m = 1$; all indices above m fail. Strict increase of C_n and comparison with C_m prove the claim. $\square$

The sequence-point floor (40) holds for C_n when $n \geq 1$, but *fails at zero*: $\lfloor \sqrt{\log_2 C_0} \rfloor = 1$. In contrast the threshold theorem correctly gives $N_C(2) = 0$.

[write] *Relation to the staircase theorem.* $N_a(y) = \max\{n : a_n \leq y\}$ is the floor-type companion of the integer staircase $N_*(y) = \min\{n : a_n \geq y\}$ of the canonical volume [14, Theorem J.43, p0:thm:staircase]: the two agree at the sequence values and $N_a = N_* - 1$ strictly between them. That theorem obtains N_* by rounding an admissible interpolation's inverse, or by a separation condition on an approximate inverse; Theorem 12.1 needs neither, because the brackets (39) leave a single candidate index and one exact comparison, the route that Chapter 8 of [13] names ("bracketing consecutive exact counts"). The paragraph after Figure 2 is the separation condition in this setting.

To locate the threshold in logarithmic coordinates, expand the square root in Proposition 10.1. With $q = 2^{-m}$,

$$\sqrt{\log_2 a_m} = m + \frac{q}{\lambda} + \left(\frac{m-1}{2\lambda} - \frac{1}{2\lambda^2 m} \right) q^2 + O(m^2 q^3). \quad (42)$$

Indeed, $L_1(m) = 2m$ contributes q/λ ; $L_2(m) = m^2 - m$ contributes $(m-1)q^2/(2\lambda)$; and the quadratic square-root term contributes $-q^2/(2\lambda^2 m)$. The next logarithmic term and all remaining square-root terms are $O(m^2 q^3)$. This is an expansion for $m \rightarrow \infty$, not a formula to be evaluated at $m = 0$.

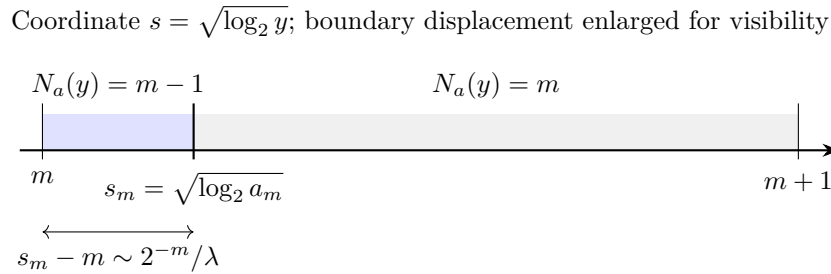


Figure 2: Within $m \leq \sqrt{\log_2 y} < m + 1$, the exact threshold is a_m . The boundary itself belongs to the right-hand interval. The diagram is schematic: the left interval has exponentially small width.

Thus $N_a(y)$ differs from m precisely when $m \leq \sqrt{\log_2 y} < \sqrt{\log_2 a_m}$. An approximate boundary may classify y only when its distance from the true comparison boundary exceeds a *proved* error bound. Inside an uncertainty interval, use the exact comparison in (41). Neither a bare big-oh statement nor flooring a truncated real inverse automatically produces an exact integer answer. Formula (42) has the same fixed-order expansion for C_m , with only the beyond-all-fixed-orders perturbation (38).

13 Exact numerical evidence and its logical role

The source packet contains exact arithmetic checks through order six; a separate fresh independent audit extends formal checking through order eight [5, 6]. The independently authored algebra

checker derives inverse coefficients using the *exponentiated* equation

$$\sum_{k \geq 0} P_k(s + \delta) t^k \exp(\lambda((2s - k)\delta + \delta^2)) = 1, \quad (43)$$

whose new linear coefficient is $2\lambda s$, and checks cancellation in both this equation and (28). This is a useful independent algebraic route, though finite-order cancellation cannot establish the infinite family of estimates proved above.

The retained independent results include:

- all 66 066 Boolean tables for $K = 1, 2, 3, 4$, with closed counts 2, 6, 48, 1194, and 44 298 direct zero-polynomial evaluations;
- all 66 067 bipartite graphs for $n = 0, 1, 2, 3, 4$, checking their weighted sums and every isolation factorial moment;
- 33 868 ordinary labeled graphs on 0 through 6 vertices, checking the colored counts by bipartiteness and connected components;
- 65 sequence indices $n = 0, \dots, 64$, 585 values of the coefficient polynomials, and 4 225 exact majorant comparisons;
- 1 012 rational inverse-threshold comparisons, including small endpoints and probes on both sides of sequence and square boundaries;
- inverse cancellation through order eight, all displayed coefficients, the three leading-coefficient identities, and agreement with all 21 source coefficient families through order six.

The majorants are checked after fourth powers so that no floating fourth root is required. Threshold floors are checked using integer bit lengths, integer square roots, and exact rational comparison, avoiding accidental rounding at boundaries. [write] The list above is the independent audit's (shipped as `code/21-fixed-audit-check_mathematics.py` with its record `data/21-fixed-audit-ev-mathematics.json`). The source packet's own checker (`code/21-fixed-src-verify_exact.py`, `data/21-fixed-src-ev-exact.json`) covers the same 66 066 tables and 66 067 graphs, 33 sequence indices $n = 0, \dots, 32$, 231 polynomial values and formal inverse cancellation through order six; its companion `code/21-fixed-src-verify_connected.py` checks the three leading-coefficient identities through order six.

A separate retained interval check encloses s_n , t_n and the signed inverse error using rational arithmetic: a positive atanh series with a geometric tail for logarithms, integer square roots for square roots, alternating Taylor bounds for e^{-x} , and outward 1024-bit dyadic rounding after arithmetic. All 36 cases with $n = 16, 32, 64$, orders $M = 1, \dots, 6$, and sequences a, C have a strictly negative upper endpoint for the inverse remainder.

For the a sequence, write

$$\mathcal{R}_{n,M} = n - s_n - \sum_{k=1}^M D_k(s_n) t_n^k, \quad J_{n,M} = -\frac{c_{M+1}}{2\lambda(M+1)!} s_n^M t_n^{M+1}.$$

Table 2 reproduces selected decimal *outward enclosures* from the retained exact endpoints. The ratio tends to one for each fixed M by Theorem 11.2; the table illustrates both approach directions and the slower onset at a larger truncation order.

n	M	Enclosure for $\mathcal{R}_{n,M}/J_{n,M}$
16	1	[1.057106335697, 1.057106335698]
32	1	[1.029841125557, 1.029841125558]
64	1	[1.015272779531, 1.015272779532]
16	3	[0.893719133773, 0.893719133774]
32	3	[0.948737227650, 0.948737227651]
64	3	[0.974858114102, 0.974858114103]
16	6	[0.532599538158, 0.532599538159]
32	6	[0.741284482732, 0.741284482733]
64	6	[0.864003938458, 0.864003938459]

Table 2: Selected rigorous inverse-error ratios from the independent audit. These are retained numerical evidence, not newly rerun computations for this article and not a substitute for the fixed-order proof.

14 Further questions and the limits of the results

Growing truncation order. The proof fixes M before letting n grow. What is the widest useful range $M = M(n)$ for which a comparably precise remainder can be established? The factor $2^{r(k-r)}$ in (2) grows rapidly with the sector index, so degree estimates alone do not settle this question. A useful answer would have to control the transition between the small- k regime and the large- k tail, rather than assert convergence of the fixed-order expansion.

[write] *Answered in part by Part II.* For the exact sector tail, Part II answers this question with all orders at once: for $n \geq 32$ and every $0 \leq M < 2n$ the tail exceeds its first omitted sector by a relative amount at most $K_n \sim 9/(13n)$, attained only at $M = 2n - 4$ (Theorem 15.1), and by at most $1/n$ for C_n with the extra table merged into the terminal sector (Theorem 15.2). Replacing the coefficient $P_k(n)$ by its leading monomial is accurate exactly when $k^2/n \rightarrow 0$ (Theorem 20.1), and with an explicit correction for $k = o(n^{2/3})$ (Theorem 21.1). The question remains open for the logarithmic and inverse expansions of Sections 10 and 11 (Section 22).

Rectangular and higher dimensional tables. For a table with p finite row labels and q finite column labels, the same boundary argument immediately gives

$$C_{p,q} = 1 + \sum_{r=0}^p \sum_{c=0}^q \binom{p}{r} \binom{q}{c} 2^{(p-r)(q-c)}.$$

This is a direct counting consequence, but asymptotics when p/q varies or one side stays bounded require their own error analysis. In dimension at least three the coordinate-flat criterion and one-positive-witness formula extend, while boundary faces overlap in a more complicated pattern. A useful next problem is an exact counting representation that organizes those overlaps without double counting.

[write] The extension to dimension $d \geq 3$ is Report 69’s “fixed-dimensional extension” [12], which states the classification for every fixed finite dimension with the same single-equation, positive-integer and unrestricted-degree qualifications.

Effective inverse certification. The finite proxy proof yields an error order and the exact threshold rule yields a final integer answer. Can explicit derivative and Taylor constants be

combined into a practical certified interval procedure that usually avoids forming the full integer a_m ? The goal would be a proved stopping criterion, with exact comparison when the interval straddles the threshold. The asymptotic expansion by itself provides no such universal rounding guarantee.

Representation cost inside the zero-auxiliary class. The polynomial (5) gives degree $2|S|$ when S is nonempty and has no all-tail cell. It need not minimize degree, coefficient height, or number of monomials. Characterizing these costs within the closed tables is separate from minimizing auxiliary arity. The present count does not answer that optimization problem.

Scope. All proofs concern fixed finite tables on positive integer inputs and one polynomial equality. All asymptotic expansions are at fixed order. Formal series manipulations are justified only by the finite truncations and error arguments given here. In particular the ordinary and exponential generating functions of a_n both have analytic radius zero, since $a_n \geq 2^{n^2}$ implies both $a_n^{1/n} \rightarrow \infty$ and $(a_n/n!)^{1/n} \rightarrow \infty$. They can still be interpreted formally. No physical simulation, source interpreter, saved schedule or theorem prover was run to prepare this article. No external submission or novelty claim is implied by its preparation.

Part II

Sharp uniform finite sector truncations of A196460

[write] Part II is manuscript 24 of batch 91, the article *Sharp uniform finite sector truncations of A196460*, author line “A companion mathematical article”, dated 4 October 2026 (12 pages as delivered). Its title block is replaced by this Part heading; its abstract and keywords follow unchanged, and its text has the label prefix `cta:us:` and the [write] notes. Its “companion article” [7] is Part I.

Abstract of Part II. The normalized terms of OEIS A196460 have an exact decomposition into $2n + 1$ positive sectors. For every $n \geq 32$, we determine the largest relative excess of a tail over its first omitted sector, simultaneously over every truncation order $0 \leq M < 2n$. The unique worst truncation is $M = 2n - 4$, its excess is an explicit rational function, and the optimal asymptotic uniform constant is $9/13$. For the clipping-table count $C_n = 1 + a_n$, merging the extra contribution into the terminal sector changes the exact maximum to $1/n$, attained uniquely at $M = 2n - 2$. Complement symmetry, a raising identity, and estimates on both discrete-Gaussian lattices give elementary proofs. Separately, a sector coefficient is asymptotic to its leading monomial precisely when $k^2/n \rightarrow 0$; an explicit central correction remains uniform for $k = o(n^{2/3})$ and yields the transition $e^{-c^2/4}$ at $k \sim c\sqrt{n}$. The finite-sector conclusions neither provide a growing-order inverse expansion nor justify replacing all earlier coefficients by their leading terms. The threshold 32 is convenient, not asserted minimal. No novelty or priority claim is made.

Keywords. A196460; finite sector expansions; sharp uniform tails; discrete Gaussian; clipping tables; growing-order asymptotics.

15 The exact sectors and the scope of the question

For an integer $n \geq 1$, let

$$a_n = \sum_{j=0}^n \binom{n}{j} (1 + 2^j)^n. \quad (44)$$

This is OEIS A196460, entered by Paul D. Hanna in 2011; the leading asymptotic $a_n \sim 2^{n^2}$ is credited there to Václav Kotesovec in 2013 [1]. Define, for $0 \leq k \leq 2n$,

$$P_k(n) = \sum_{\substack{r+c=k \\ 0 \leq r, c \leq n}} \binom{n}{r} \binom{n}{c} 2^{rc}, \quad S(n, k) = P_k(n) 2^{-nk}, \quad A_n = \frac{a_n}{2^{n^2}}. \quad (45)$$

Each $P_k(n)$ is strictly positive: every allowed k admits at least one pair (r, c) in the indicated square. Expanding (44), then replacing both binomial indices by their complements, gives

$$\begin{aligned} a_n &= \sum_{u,v=0}^n \binom{n}{u} \binom{n}{v} 2^{uv} = \sum_{r,c=0}^n \binom{n}{r} \binom{n}{c} 2^{(n-r)(n-c)}, \\ A_n &= \sum_{r,c=0}^n \binom{n}{r} \binom{n}{c} 2^{-n(r+c)+rc} = \sum_{k=0}^{2n} S(n, k). \end{aligned} \quad (46)$$

Thus the sector decomposition is finite and exact, including both endpoints.

[write] This is Part I's identity (11), re-derived; A_n is Part I's $\mathcal{A}_n$, and $P_k(n)$ is Part I's polynomial (2) at the integer n .

The companion article [7] proves the zero-auxiliary-variable classification for Boolean tables clipped at $K = n + 1$, as well as the count $C_n = 1 + a_n$. Briefly, an accepted boundary mark forces its full row or column; with the tail corner rejected, r row marks and c column marks leave $(n-r)(n-c)$ free interior entries. Summing their choices gives the double sum in (46). The accepted corner forces the single full table, accounting for the extra 1. We use this established count without repeating its closure classification or inverse theory.

[write] The companion is Part I of this report: the classification is Section 6 (Report 69's theorem, [12]), the count Section 7, the graph interpretation Section 8, and the inverse theory Sections 10–12.

The same companion explains the graph interpretation $a_n = \sum_G 2^{i(G)}$: G ranges over graphs on two distinguished labeled parts of size n , and $i(G)$ counts isolated vertices. Choosing a subset of isolated vertices marks the boundary. Its sector interpretation is $S(n, k) = \mathbb{E}(\binom{i(G)}{k})$ for the uniform random graph. This is consistent with the existing two-unary, one-binary relational model for A196460 in Svatoš et al. [2]; the two graph parts are distinct copies of the domain, so all n^2 edges, including pairs of corresponding labels, remain available. These interpretations are credited background, not claims of a new model.

For every allowed truncation order, put

$$R_{n,M} = \sum_{k=M+1}^{2n} S(n, k), \quad E_{n,M} = \frac{R_{n,M}}{S(n, M+1)} - 1, \quad 0 \leq M < 2n. \quad (47)$$

The question here is stronger than a statement for each fixed M : how well does the first omitted *exact* sector approximate the tail, uniformly over all $2n$ possible values of M ?

Theorem 15.1 (Exact sharp bound for every truncation). *For every integer $n \geq 32$, simultaneously for all integers $0 \leq M < 2n$,*

$$0 \leq E_{n,M} \leq K_n, \quad K_n = \frac{3(3n^2 + n + 1)}{n(13n^2 - 15n + 2)}. \quad (48)$$

The upper equality holds uniquely at $M = 2n - 4$. Moreover,

$$K_n = \frac{9}{13n} + \frac{174n + 21}{13n(13n^2 - 15n + 2)}, \quad \lim_{n \rightarrow \infty} \max_{0 \leq M < 2n} nE_{n,M} = \frac{9}{13}. \quad (49)$$

In particular $9/13$ is the optimal asymptotic uniform constant.

Theorem 15.2 (Sharp bound for the clipping-table count). *Let $C_n = a_n + 1$, and merge its additional normalized contribution into the terminal sector only:*

$$S_C(n, k) = \begin{cases} S(n, k), & 0 \leq k < 2n, \\ 2S(n, 2n), & k = 2n. \end{cases} \quad \frac{C_n}{2n^2} = \sum_{k=0}^{2n} S_C(n, k). \quad (50)$$

For $n \geq 32$, simultaneously for $0 \leq M < 2n$,

$$0 \leq E_{n,M}^C := \frac{\sum_{k=M+1}^{2n} S_C(n, k)}{S_C(n, M+1)} - 1 \leq \frac{1}{n}. \quad (51)$$

The upper equality holds uniquely at $M = 2n - 2$. At $M = 2n - 1$ the relative excess is zero. The sharp asymptotic uniform constant in this merged convention is therefore 1.

The threshold 32 in both statements is chosen for the elementary estimates below. No minimal-threshold assertion is intended. Also, the normalization in Theorem 15.2 matters: if the additional 2^{-n^2} is left outside the sectors and the old denominator S is used, the excess at $M = 2n - 1$ is exactly 1, not $O(1/n)$.

16 Complement symmetry and the raising identity

Two exact identities connect the middle of the sector range with its ends. First, substituting $r = n - u$, $c = n - v$ gives

$$-n(r + c) + rc = -n^2 + uv.$$

Binomial symmetry therefore proves

$$S(n, 2n - k) = 2^{-n^2} P_k(n), \quad 0 \leq k \leq 2n. \quad (52)$$

In particular $P_0(n) = 1$ and $S(n, 2n) = 2^{-n^2}$, which explains the terminal merge in (50).

Write $w_{r,c} = \binom{n}{r} \binom{n}{c} 2^{rc}$, with all sums below restricted to valid r, c . For $0 \leq k < 2n$, the raising identity is

$$(k+1)P_{k+1}(n) = \sum_{r+c=k} w_{r,c} [(n-r)2^c + (n-c)2^r]. \quad (53)$$

Indeed, $(n-r)\binom{n}{r} = (r+1)\binom{n}{r+1}$, and multiplication by 2^c changes 2^{rc} to $2^{(r+1)c}$. Raising the first coordinate therefore gives a target weight times its new first coordinate; raising the second gives that weight times its new second coordinate. Every target of total degree $k+1$ is counted with total

multiplier $k + 1$. Forbidden boundary moves have factor $n - r = 0$ or $n - c = 0$. This argument also covers $k = 0$ and $k = 2n - 1$.

Set

$$\lambda = \log 2, \quad h = r - k/2, \quad D = n - k/2.$$

The bracket in (53) equals

$$2^{k/2} [2D \cosh(\lambda h) + 2h \sinh(\lambda h)]. \quad (54)$$

Here $D > 0$ for $k < 2n$, $\cosh(\lambda h) \geq 1$, and $h \sinh(\lambda h) \geq 0$. Dividing the sum by the positive $P_k(n)$ proves

$$\frac{P_{k+1}(n)}{P_k(n)} \geq \frac{2n - k}{k + 1} 2^{k/2}, \quad 0 \leq k < 2n. \quad (55)$$

For a high-sector ratio, put $\ell = 2n - k$. If $n \leq k < 2n$, then $1 \leq \ell \leq n$; complement symmetry and (55) at $\ell - 1$ give

$$\frac{S(n, k + 1)}{S(n, k)} = \frac{P_{\ell-1}(n)}{P_\ell(n)} \leq g_{n,\ell} := \frac{\ell 2^{-(\ell-1)/2}}{2n - \ell + 1} \leq q_n := \frac{3}{2(n + 1)}. \quad (56)$$

For the last inequality, the numerator has values $1, \sqrt{2}, 3/2$ at $\ell = 1, 2, 3$, and thereafter its next-to-current ratio is $(\ell + 1)/(\sqrt{2}\ell) < 1$. Its exact maximum is thus $3/2$. The denominator is at least $n + 1$ throughout $1 \leq \ell \leq n$.

17 Both discrete Gaussian lattices and every adjacent ratio

To control low sectors we need an upper, rather than a lower, bound in (53). At fixed k , its normalized summand weights in h are proportional to

$$b(h)2^{-h^2}, \quad b(h) = \binom{n}{k/2 + h} \binom{n}{k/2 - h}. \quad (57)$$

The lattice is $\mathbb{Z}$ for even k and $\mathbb{Z} + \frac{1}{2}$ for odd k . The factor $2^{k^2/4}$ cancels on normalizing. Extend b by zero outside the allowed finite support. It is symmetric, and for nonnegative h where the ratio is defined,

$$\frac{b(h + 1)}{b(h)} = \frac{(D - h)(k/2 - h)}{(k/2 + h + 1)(D + h + 1)} \leq 1. \quad (58)$$

At the edge of the support the next value is zero, and beyond that b remains zero. Thus b decreases with $|h|$, including the degenerate supports at $k = 0, 1$.

Lemma 17.1 (Decreasing reweighting). *On either of the two lattices, give h probability proportional to 2^{-h^2} . If b is nonnegative, symmetric, decreasing with $|h|$, and has positive mean, then for every integrable nonnegative increasing function f of $|h|$,*

$$\mathbb{E}_b f := \frac{\mathbb{E}[b(h)f(|h|)]}{\mathbb{E}b(h)} \leq \mathbb{E}f(|h|), \quad (59)$$

provided the displayed expectations exist.

Proof. For independent variables X, Y with the base Gaussian distribution,

$$(b(X) - b(Y))(f(|X|) - f(|Y|)) \leq 0.$$

Its expectation is $2 \operatorname{Cov}(b, f)$. Dividing the resulting inequality by $\mathbb{E}b > 0$ proves the claim. We apply the argument separately on each lattice, never by comparing the two lattices to each other. $\square$

Take

$$f(x) = \cosh(\lambda x) + x \sinh(\lambda x), \quad x \geq 0.$$

Its derivative is nonnegative, and $f(x) \leq (1+x)2^x$. If $n \geq 2$ and $0 \leq k \leq n$, then $D \geq 1$. The expression in (54), divided by $2D2^{k/2}$, is therefore at most $f(|h|)$. We give an explicit common bound of 8 for its expectation.

On the integer lattice the Gaussian normalizing sum is at least 1. The $j = 0, 1, 2$ terms of the majorant sum to $13/2$. Since $j(j-1) \geq 2j$ for $j \geq 3$,

$$\begin{aligned} \mathbb{E}_{\mathbb{Z}} f(|h|) &\leq 1 + 2 \sum_{j \geq 1} (j+1) 2^{-j(j-1)} \\ &\leq \frac{13}{2} + 2 \sum_{j \geq 3} (j+1) 4^{-j} = \frac{481}{72} < 8. \end{aligned} \quad (60)$$

On the half-integer lattice the two points $\pm \frac{1}{2}$ give a normalizing sum at least $2 \cdot 2^{-1/4}$. Set $h = m + \frac{1}{2}$ on the nonnegative half. Then

$$\begin{aligned} \mathbb{E}_{\mathbb{Z}+1/2} f(|h|) &\leq \sqrt{2} \sum_{m \geq 0} (m + \tfrac{3}{2}) 2^{-m^2} \\ &\leq \sqrt{2} \left[\frac{3}{2} + \sum_{m \geq 1} (m + \tfrac{3}{2}) 2^{-m} \right] = 5\sqrt{2} < 8. \end{aligned} \quad (61)$$

The last step uses $m^2 \geq m$ for $m \geq 1$. These majorants also verify the absolute convergence required in Lemma 17.1.

Averaging the raising bracket and applying that lemma now gives

$$\frac{P_{k+1}(n)}{P_k(n)} \leq 8 \frac{2n-k}{k+1} 2^{k/2}, \quad \frac{S(n, k+1)}{S(n, k)} \leq d_n := 16n 2^{-n/2}, \quad 0 \leq k \leq n. \quad (62)$$

The second bound deliberately sacrifices sharpness: $2n - k \leq 2n$, $k+1 \geq 1$, and $k/2 - n \leq -n/2$ suffice.

At $n = 32$, $d_n = 1/128$ and $(n+1)d_n = 33/128 < 3/2$. For successive $n \geq 32$, the ratio of $(n+1)d_n$ is $(n+2)/(\sqrt{2}n) < 1$. Hence $d_n \leq q_n$ for every $n \geq 32$. The low range $0 \leq k \leq n$ and the high range $n \leq k < 2n$ overlap at $k = n$ and leave no gap. We have proved

$$\frac{S(n, k+1)}{S(n, k)} \leq q_n < 1 \quad \text{for all } 0 \leq k < 2n, \quad n \geq 32. \quad (63)$$

Thus the S sectors decrease strictly, and (52) makes the P sectors increase strictly. A finite geometric majorant already gives the useful preliminary estimate

$$0 \leq E_{n,M} \leq \frac{q_n}{1 - q_n} = \frac{3}{2n-1}, \quad 0 \leq M < 2n. \quad (64)$$

All-order uniformity has therefore been established before any attempt to locate the sharp maximum.

18 The unique worst tail

Put $j = M + 1$ and $\ell = 2n - j$. Then $1 \leq j \leq 2n$ and $0 \leq \ell \leq 2n - 1$. Reversing the finite tail using (52) yields the exact identity

$$E_{n,M} = H_\ell, \quad H_0 = 0, \quad H_\ell = \frac{P_0 + \cdots + P_{\ell-1}}{P_\ell} \quad (\ell \geq 1), \quad (65)$$

where the common argument n is suppressed. Notice that $\ell = 2n$ is absent: it would correspond to $M = -1$, retaining no sectors. Equation (63) also implies

$$H_\ell \leq \frac{P_{\ell-1}/P_\ell}{1 - q_n}. \quad (66)$$

We separate the possible ℓ into three ranges.

18.1 Low first omitted sectors

If $j < n$, the first ratio in the tail is bounded by d_n , while every later ratio is bounded by q_n . Consequently $H_\ell \leq d_n/(1 - q_n)$. The quantity $nd_n/(1 - q_n) = 16n^2 2^{-n/2}/(1 - q_n)$ decreases for $n \geq 32$: $n^2 2^{-n/2}$ decreases because $((n+1)/n)^4 < 2$, and $1/(1 - q_n)$ also decreases. At $n = 32$ their product is $11/42$. Thus

$$nH_\ell \leq \frac{11}{42} < \frac{9}{13} \quad (j < n). \quad (67)$$

18.2 High first omitted sectors away from the endpoint

If $j \geq n$, then $0 \leq \ell \leq n$. For $5 \leq \ell < n$,

$$\frac{g_{n,\ell+1}}{g_{n,\ell}} = \frac{\ell+1}{\ell} \frac{2n-\ell+1}{2n-\ell} \frac{1}{\sqrt{2}} \leq \frac{6}{5} \frac{34}{33} \frac{1}{\sqrt{2}} < 1. \quad (68)$$

Indeed, $2n - \ell \geq n + 1 \geq 33$. Therefore for $5 \leq \ell \leq n$,

$$g_{n,\ell} \leq g_{n,5} = \frac{5}{8(n-2)}, \quad H_\ell \leq B_n := \frac{5(n+1)}{4(n-2)(2n-1)}. \quad (69)$$

We will compare this bound directly with H_3 .

18.3 The four small nonzero endpoint distances

The finite definition gives

$$\begin{aligned} P_0 &= 1, & P_1 &= 2n, & P_2 &= 3n^2 - n, \\ P_3 &= \frac{13n^3 - 15n^2 + 2n}{3}, & P_4 &= \frac{27n^4 - 66n^3 + 41n^2 - 2n}{4}. \end{aligned} \quad (70)$$

For example, $P_3 = 2\binom{n}{3} + 8n\binom{n}{2}$, and $P_4 = 2\binom{n}{4} + 16n\binom{n}{3} + 16\binom{n}{2}^2$; these also make the formulas directly checkable. [write] These are Part I's (13) at $x = n$. Thus

$$\begin{aligned} H_1 &= \frac{1}{2n}, & H_2 &= \frac{2n+1}{n(3n-1)}, \\ H_3 &= \frac{3(3n^2 + n + 1)}{n(13n^2 - 15n + 2)}, & H_4 &= \frac{4(13n^3 - 6n^2 + 5n + 3)}{3n(27n^3 - 66n^2 + 41n - 2)}. \end{aligned} \quad (71)$$

All denominators are positive for the present range, either directly or from the positive sector definitions. In particular, $13n^2 - 15n + 2 = (13n - 2)(n - 1) > 0$ for $n \geq 2$.

Algebra gives (49) with $K_n = H_3$, so $nH_3 > 9/13$. In contrast,

$$nH_1 = \frac{1}{2} < \frac{9}{13}, \quad nH_2 < \frac{9}{13} \quad (n > 22).$$

For H_4 , the comparison $nH_4 < 9/13$ reduces, after multiplication by positive denominators, to

$$53n^3 - 1470n^2 + 847n - 210 = n^2(53n - 1470) + 847n - 210 > 0 \quad (n \geq 28). \quad (72)$$

Finally, $B_n < H_3$ in (69) reduces to

$$7n^4 - 146n^3 + 101n^2 - 46n + 24 > 0. \quad (73)$$

This holds already for $n \geq 21$, since $n^3(7n - 146) > 0$ and $101n^2 - 46n + 24 > 0$ there.

The strict comparisons cover all allowed indices: $H_0 = 0$; $\ell = 1, 2, 4$ are below $9/(13n)$; $\ell = 3$ is above it; $5 \leq \ell \leq n$ are below H_3 ; and $j < n$ is covered by (67). This proves Theorem 15.1, including uniqueness. The change of index takes $\ell = 3$ precisely to $M = 2n - 4$. Because the maximum is computed exactly for each $n \geq 32$, its limiting constant is the ordinary rational-function limit in (49); no interchange of an uncontrolled supremum and a limit is involved. No smaller asymptotic uniform constant can work at the maximizing truncation.

19 The extra full table and the terminal endpoint

The added 1 in $C_n = a_n + 1$ has normalized size $2^{-n^2} = S(n, 2n)$. For $\ell \geq 1$ the first omitted sector is not terminal and is unchanged by merging. Dividing the added term by that sector gives

$$E_{n,M}^C = J_\ell := H_\ell + \frac{1}{P_\ell} \quad (\ell \geq 1), \quad J_0 = 0. \quad (74)$$

The separate value at $\ell = 0$ is essential: the denominator itself is then the doubled terminal sector, and the tail contains only that sector. Formula $H_\ell + 1/P_\ell$ must not be used at $\ell = 0$.

Since $P_1 = 2n$, $J_1 = 1/n$. For $\ell \geq 2$, strict increase of the P sectors and Theorem 15.1 give

$$J_\ell \leq H_3 + \frac{1}{P_2} < \frac{1}{n}.$$

The last strict inequality is exactly, after clearing positive denominators,

$$12n^3 - 71n^2 + 30n - 1 > 0. \quad (75)$$

For $n \geq 6$, its first two terms equal $n^2(12n - 71) > 0$, and $30n - 1 > 0$ as well. This proves Theorem 15.2, with unique maximizer $\ell = 1$, or $M = 2n - 2$.

If one retains the unmerged S sectors for C_n , the same calculation still applies for $M \leq 2n - 2$. At the last truncation, however,

$$\frac{C_n}{2^{n^2}} - \sum_{k=0}^{2n-1} S(n, k) = 2S(n, 2n).$$

n	maximizer ℓ for a_n	$\max E_{n,M}$	maximizer ℓ for merged C_n	$\max E_{n,M}^C$
1	1	1/2	1	1
2	3	15/16	3	1
5	9	2817473/10485760	9	1408737/5242880
6	4	35/222	1	1/6
8	4	2105/19852	1	1/8
9	3	11/120	1	1/9
32	3	45/1984	1	1/32
64	3	12353/1115520	1	1/64
128	3	49281/9005824	1	1/128
160	3	76961/17621440	1	1/160

Table 3: Exact tail maxima transcribed from the independent audit’s retained data [11]. Each displayed maximizer is unique. The original truncation index is $M = 2n - \ell - 1$. The rows below 32 illustrate finite behavior only; they do not extend the proved all- n threshold. No checker was rerun for this article.

Relative to the old first omitted sector, the excess is exactly 1. One may instead retain the additional term separately and approximate the a_n part by Theorem 15.1; discarding it is the operation that fails at this endpoint.

[write] At placement (4 October 2026) the source and audit checkers of this Part were rerun on copies and passed, with outputs equal to the retained ones up to line endings, and the rows $n = 1, 2, 5, 6, 8, 9$ and the maxima at $n = 32, 33, 40, 64$ were recomputed by fresh code; see the report README.

20 Leading monomials have a different uniformity boundary

The exact sectors are available over the full finite range, but replacing $P_k(n)$ by its leading monomial is a different approximation. Define

$$\alpha_k = \sum_{r=0}^k \frac{2^{r(k-r)}}{r!(k-r)!} > 0. \quad (76)$$

For fixed k , the polynomial obtained from the binomial products has degree k and leading coefficient α_k . The integer sequence $k!\alpha_k = \sum_r \binom{k}{r} 2^{r(k-r)}$ is the established count of labeled graphs with two named proper colors, OEIS A047863 [8]. Its connected counterpart for $k \geq 1$ is A002027 [9]; the entry’s value at zero is a separate convention. These known graph counts are background to the fixed-order logarithmic theory, not additional results needed for the tail maximization.

[write] α_k is Part I’s (12), and $k!\alpha_k$ and its connected counterpart are Part I’s b_k and c_k (Section 10).

For $0 \leq k \leq n$ and $r + c = k$, write

$$B_{n,r,c} = \prod_{i=0}^{r-1} \left(1 - \frac{i}{n}\right) \prod_{i=0}^{c-1} \left(1 - \frac{i}{n}\right). \quad (77)$$

Then $P_k(n)/(\alpha_k n^k)$ is the weighted mean of these products with positive weights proportional to $2^{rc}/(r!c!)$. All factors belong to $(0, 1]$. The elementary inequality $\prod_i (1 - x_i) \geq 1 - \sum_i x_i$ for

$0 \leq x_i \leq 1$ yields

$$1 - \frac{k(k-1)}{2n} \leq \frac{P_k(n)}{\alpha_k n^k} \leq 1, \quad 0 \leq k \leq n. \quad (78)$$

Here $r(r-1) + c(c-1) \leq k(k-1)$; a negative lower bound causes no problem.

There is also a useful upper bound on the entire finite range:

$$\frac{P_k(n)}{\alpha_k n^k} \leq \exp\left(-\frac{k(k-2)}{4n}\right), \quad 0 \leq k \leq 2n. \quad (79)$$

For a valid pair $r, c \leq n$, every product factor is positive, including when $r = n$ or $c = n$. With $h = r - k/2$, use $\log(1-x) \leq -x$ and

$$r(r-1) + c(c-1) = \frac{k(k-2)}{2} + 2h^2$$

to bound $B_{n,r,c}$ by $\exp[-k(k-2)/(4n) - h^2/n]$. When $k > n$, the invalid pairs in the full α_k sum contribute zero to $P_k(n)$ and obey the same upper bound. Averaging proves (79). At $k = 1$ the displayed exponent is positive, which is harmless.

Theorem 20.1 (Exact boundary for leading monomials). *For any integer sequence $0 \leq k(n) \leq 2n$,*

$$\frac{P_{k(n)}(n)}{\alpha_{k(n)} n^{k(n)}} \rightarrow 1 \iff \frac{k(n)^2}{n} \rightarrow 0. \quad (80)$$

Proof. If $k^2/n \rightarrow 0$, then eventually $k \leq n$, and (78) proves convergence. Conversely, if k^2/n does not tend to zero, some subsequence satisfies $k^2/n \geq \delta > 0$. On this subsequence $k \rightarrow \infty$, so eventually $k(k-2)/(4n) \geq \delta/8$. Equation (79) then bounds the ratio above by $e^{-\delta/8} < 1$. This contradicts convergence to 1. No monotonicity of $k(n)$ is assumed. $\square$

21 An explicit central correction

The square-root scale is the boundary for the uncorrected leading term. A simple correction gives a wider uniform range.

Theorem 21.1 (Corrected coefficient estimate). *For $n \geq 3$ and $0 \leq k \leq n$, put*

$$U = \frac{k(k-2)}{4n}, \quad \varepsilon = \frac{k(k-1)(2k-1)}{12n^2(1-(k-1)/n)}. \quad (81)$$

Then

$$e^{-\varepsilon} \left(1 - \frac{2}{n}\right) \leq e^U \frac{P_k(n)}{\alpha_k n^k} \leq 1. \quad (82)$$

In particular the corrected ratio differs from 1 by at most $\varepsilon + 2/n$. On the explicit subrange $0 \leq k \leq n/2$,

$$0 \leq 1 - e^U \frac{P_k(n)}{\alpha_k n^k} \leq \frac{2}{n} + \frac{k^3}{3n^2}. \quad (83)$$

Consequently, uniformly for $k = o(n^{2/3})$,

$$\frac{P_k(n)}{\alpha_k n^k} = \exp\left(-\frac{k(k-2)}{4n}\right) \left[1 + O\left(\frac{1}{n} + \frac{k^3}{n^2}\right)\right]. \quad (84)$$

If $k/\sqrt{n} \rightarrow c \in (0, \infty)$, then

$$\frac{P_k(n)}{\alpha_k n^k} \rightarrow e^{-c^2/4}. \quad (85)$$

Proof. The α_k weights of $h = r - k/2$ are proportional to

$$\binom{k}{k/2 + h} 2^{-h^2}.$$

The binomial factor is symmetric and decreases with $|h|$ on the appropriate parity lattice. Applying Lemma 17.1 to $f(|h|) = h^2$ bounds its second moment by the base Gaussian second moment. On the integer lattice, with denominator at least 1,

$$\mathbb{E}_{\mathbb{Z}} h^2 \leq 1 + 2 \sum_{j \geq 2} j^2 4^{-j} = \frac{107}{54} < 2. \quad (86)$$

The $j = 1$ terms give 1, and $j^2 \geq 2j$ for $j \geq 2$ justifies the majorant. On the half-integer lattice, divide by the lower normalizing bound $2 \cdot 2^{-1/4}$ and use $m(m+1) \geq 2m$ to obtain

$$\mathbb{E}_{\mathbb{Z}+1/2} h^2 \leq \sum_{m \geq 0} (m + \tfrac{1}{2})^2 4^{-m} = \frac{41}{27} < 2. \quad (87)$$

Thus in the α_k distribution, $\mathbb{E} h^2 < 2$ on either lattice.

For $0 \leq x < 1$, the series for $-\log(1-x)$ gives

$$-x - \frac{x^2}{2(1-x)} \leq \log(1-x) \leq -x, \quad (88)$$

because $\sum_{m \geq 2} x^m/m \leq x^2/[2(1-x)]$. Apply this to the product (77). Every denominator $1 - i/n$ is at least $1 - (k-1)/n > 0$. Also

$$\sum_{i=0}^{r-1} i^2 + \sum_{i=0}^{c-1} i^2 \leq \sum_{i=0}^{k-1} i^2 = \frac{k(k-1)(2k-1)}{6}.$$

To see the inequality, shift the second block of indices from $0, \dots, c-1$ to $r, \dots, r+c-1$, which can only increase their squares. Summing (88) therefore gives

$$-U - \frac{h^2}{n} - \varepsilon \leq \log B_{n,r,c} \leq -U - \frac{h^2}{n}. \quad (89)$$

Exponentiate and average in the α_k distribution. Since $e^{-h^2/n} \geq 1 - h^2/n$,

$$e^{-\varepsilon}(1 - 2/n) \leq e^{-\varepsilon} \mathbb{E} e^{-h^2/n} \leq e^U \frac{P_k(n)}{\alpha_k n^k} \leq \mathbb{E} e^{-h^2/n} \leq 1.$$

This proves (82). The cases $k = 0, 1$ are also included; their products may be checked directly and $\varepsilon = 0$. At $k = n$ the denominator in (81) is $1/n > 0$, so that endpoint remains valid even though the numerical bound is loose.

Using $1 - e^{-\varepsilon} \leq \varepsilon$ bounds the deficit by $\varepsilon + 2/n$. For $k \leq n/2$, $1 - (k-1)/n \geq 1/2$ and $k(k-1)(2k-1) \leq 2k^3$, giving $\varepsilon \leq k^3/(3n^2)$ and (83). Every sequence $k = o(n^{2/3})$ eventually lies in this subrange and has vanishing error, proving the uniform meaning of (84). Finally, if $k/\sqrt{n} \rightarrow c$, then $U \rightarrow c^2/4$ and the corrected ratio tends to 1, which yields (85). $\square$

22 What the uniformity does and does not provide

Theorems 15.1 and 15.2 control a finite sum of exact positive sectors. They permit any integer truncation order $M = M(n)$ in the full range $0 \leq M < 2n$, even one approaching the terminal endpoint. The leading-monomial statements instead compare one coefficient at a time. Their narrower ranges are compatible with, and do not limit, the exact-sector tail theorem.

There is an additional distinction between coefficientwise relative accuracy and remainder-scale accuracy. Replacing every retained $P_k(n)$ by $\alpha_k n^k$ can introduce errors in earlier sectors that are much larger than the first omitted sector. Neither Theorem 20.1 nor Theorem 21.1 licenses that substitution at the tail’s relative-error scale. The precise bounds proved here always identify which quantity is being approximated.

Similarly, a uniform finite-sector estimate does not by itself extend the companion’s fixed-order logarithmic or Laurent inverse expansions to growing orders. Composition, logarithms, and inversion require control of their own coefficients, accumulated errors, and any derivatives used in a real-variable argument. No convergence of an infinite sector series at noninteger arguments, and no exact integer inverse obtained by flooring an approximate real inverse, is claimed.

Several further questions are suggested by these proofs, but are not settled here. A growing-order logarithmic or inverse theorem would need an explicit admissible order range and an error bound stable under the relevant transformations. A central-range extension of (82) would require higher-order product corrections with uniform control beyond $k = o(n^{2/3})$. Parity-sensitive Gaussian or theta refinements are a natural tool for that comparison, but the underlying theta asymptotic for $k!\alpha_k$ is already recorded in A047863, credited to Kotesovec in 2013 [8]. The classical colored-graph literature is also identified there and in A002027. The remaining question is the additional uniformity for this two-parameter coefficient and its use in transformed expansions, not the discovery of the known parity constants.

[write] Part I’s question “Growing truncation order” (Section 14) is thereby answered for the exact sector tail and stays open, as stated here, for the logarithmic and inverse expansions; the questions of this section are open.

A Evidence provenance and preservation boundary (Part I)

[write] This appendix is manuscript 21’s. The packets it names ship in this report’s directory under the prefix 21-fixed-: the source packet (delivered `inputs/asymptotics`) as `21-fixed-src-*`, the independent audit (`inputs/independent-audit`) as `21-fixed-audit-*`; the “accompanying source-and-evidence archive” survives in the arrival commit (Appendix C).

The mathematical source is the sealed packet `arity-table-asymptotics-20261004`; the fresh independent audit is `independent-arity-asymptotics-audit-20261004`. The source and audit are retained in full in the accompanying source-and-evidence archive. Their code is historical evidence and is inert during the presentation build. Only separately owned presentation and packaging tools are used to compile and render this article. The accepted one-witness proof and its earlier audit are also retained as pinned text dependencies; the classification needed here is proved in full in Section 6.

Object	SHA-256
Source manifest	6ae30398c52b6fe1d07a91346791ed3355 e7104c55b19a4103de257299293de1
Source proof	638e524d058deb926919d15b7df2dafc17 f7a44dc2688b33e06ee24f476b7c41
Independent audit manifest	737af333985bc079ecdd688f1854f7221 bb86169db7841d91a1115d2f94e7055
Independent audit text	7405fc28ea33b90ca80471ab368b667e 1728b374eb3098f3e62844a91e1f8ca7

The source’s authentic historical inventories recorded 249 objects before and 376 in its later inventory, with exactly 132 differences: 44 in the then-active Report 69 release tree and 88 in the Report 70 tree. These historical changes are preserved as recorded, not replaced by a retrospectively cleaner snapshot. The 62 core objects outside those concurrent release trees were unchanged in the recorded interval, including bytes, sizes, modes, nanosecond modification times and directory metadata. The fresh audit independently recomputed that comparison. It also established its own equality boundary over 87 specified input objects, beginning at 2026-10-04 15:23:38 UTC. Access times were excluded, because reads may update them. Neither historical result asserts whole-interval preservation of the two active release trees or attributes their changes to a particular actor.

The article packaging records a separate, scoped before-and-after inventory for the actual source and audit inputs it copies. Source bytes, file modes and modification times are preserved in the copied dossier; the archive’s inventory and verifier specify how full metadata is retained beyond the coarse timestamp field of a ZIP entry. Exact external build dependencies are locked separately from the mathematical inputs. The presentation build uses an isolated output directory, no shell escape, and full-page rendering. These are reproducibility and preservation boundaries, not claims of operating-system WORM storage or independent time attestation.

The public OEIS page and versioned arXiv record/PDF text were consulted on 4 October 2026. The A196460 row was checked in extracted PDF text at printed page 32; a web screenshot request failed, so no visual inspection of that remote page is claimed. No raw arXiv PDF byte pin or independent authentication of its remote revision history is asserted. The source’s bounded literature and workspace searches are retained with their limitations; they do not establish originality. The present article does not repeat those searches as an exhaustive negative result.

B Evidence and preservation (Part II)

[write] This appendix is manuscript 24’s. Its source packet (delivered `inputs/uniform-sectors`) ships as `24-unif-src-*`, its audit (`inputs/independent-audit`) as `24-unif-audit-*`; its “predecessor article release” is manuscript 21, shipped once, as Part I’s files.

The mathematical source packet [10] and its fresh independent audit [11] are retained in full in the source-evidence bundle. The audit’s verdict is that no mathematical correction is required. The proofs in this article are analytic; finite exact computations corroborate their identities and comparisons but do not replace the all- n arguments.

The source’s stored standard-library checks cover $n = 1, \dots, 128$: 16 640 complement identities, 16 512 raising identities and squared lower bounds, 15 520 global adjacent-ratio comparisons, 97 complete maxima checks for each tail convention, and 2 498 elementary leading-sector comparisons. Its algebra evidence records the strict polynomial comparisons, Gaussian sums, and 129 exact

leading-weight second moments.

The independent audit uses newly authored integer and rational arithmetic over $n = 1, \dots, 160$. It records 320 direct normalizations, 25 920 complement identities, 25 760 raising identities and squared lower bounds, and 24 768 global-ratio comparisons. It checks each sharp tail inequality, including equality only at its claimed unique index, in 24 768 cases per convention. It also records 13 040 leading-product comparisons, 4 272 rigorous corrected-coefficient comparisons using rational exponential enclosures, and 321 exact second-moment comparisons. The corrected finite test range is only a corroborating subset; the analytic proof covers all $n \geq 3$, $0 \leq k \leq n$.

No scientific program in either packet or an upstream dependency was run or imported in preparing this article. The stored programs and their historical output are inert evidence. Only separately inspected presentation and packaging tools are used for typesetting, rendering, reproducibility checks, and archiving. Table 3 is a transcription of retained exact data, not a new scientific computation.

The source records an unchanged 51-object input boundary, captured after initial inert inspection and before its new sector checker ran. The independent audit records a separate unchanged 333-object boundary, including the full predecessor article release and its specified adjacent artifacts. These scoped historical claims are preserved as recorded. The source’s narrower boundary does not assert preservation of unrelated concurrent article trees; the audit’s later snapshot does not retroactively broaden that earlier claim. Earlier dependency histories retain their own qualifications.

The presentation bundle records its own before-and-after boundary for the inputs it reads and copies. It preserves source bytes, permission modes, and nanosecond modification times, including directory metadata; access times are excluded because reading may update them. The predecessor article is not revised. Authenticated copying and extraction record metadata beyond ZIP’s coarse timestamp field. External build dependencies are separately locked, and an isolated fresh-format no-shell-escape build records all three typesetting passes and their input union. The final PDF is rendered page by page for visual review. Read-only sealing and digest checks are integrity measures, not WORM storage or an independently trusted timestamp.

Object	SHA-256
Source proof	ebd92e3b5bdf684981b2ff248376ac29 0c57aec403169beee780fef9ea92aa65
Source manifest	bf48296dc1bd86689df3dbef9cb3bba99 903ec84a32f226e2289ec79ea0a3a29
Independent audit	4fa84b2ddfe383668c86ea0ad4d7419b 0304eb0d9a97725aecbdab7e3b63ff03
Audit manifest	039c112bd070a8b48b92584c76ec05b3 876662bb82f186813a0fb445ddd9e36c

The public A196460, A047863, and A002027 pages were consulted as text on 4 October 2026 for the precise sequence attributions and known graph and theta context. The relational-model citation is also documented in the predecessor article’s retained primary-source check. No new exhaustive literature search or priority audit was performed. Neither the source archive nor this article is a theorem-prover verification or an endorsement of every unrelated scientific object retained in upstream dependency archives.

C Provenance of this report and editorial notes

[write] Written 4 October 2026 (batch 91O).

Sources. The report was built from two manuscripts, numbers 21 and 24 of the 24 archives of batch 91 of ProveIt’s incoming-reports intake, and the last two to be placed. Both arrived in commit `0d7f51c44` and were placed in this report’s directory by commit `ec8ae3dc7`, which also removed both archives from `docs/incoming`; they survive in the arrival commit (`git show 0d7f51c44:docs/incoming/<archive>`).

- *Manuscript 21* (Part I, the base): archive `A196460_asymptotics_and_inversion_sources.zip` (4,852,913 bytes, 176 files, package directory `ArityAsymptotics/`); main file `article.tex` (1,046 lines, 65 labels, SHA-256 `6569cdc9...`), a 16-page PDF (`2a95d839...`), titled *Zero auxiliary clipping tables: Exact counts and all orders asymptotics of A196460*, author line “A standalone mathematical research article”, dated 4 October 2026.
- *Manuscript 24* (Part II): archive `A196460_sharp_uniform_truncations_sources.zip` (17,225,521 bytes, 584 files, package directory `UniformSectors/`); main file `article.tex` (811 lines, 61 labels, `d0770c2f...`), a 12-page PDF (`e1689dc2...`), titled *Sharp uniform finite sector truncations of A196460*, author line “A companion mathematical article”, dated 4 October 2026.

Pins. Neither manuscript names a ProveIt commit. Manuscript 21’s source notes record that three public ProveIt web pages were fetched on 4 October 2026 and contained no A196460, with no remote commit pin. Manuscript 24 pins manuscript 21 by the SHA-256 of its delivered archive, `821a1cfc1c00fd6d...` (its `INPUT_PINS.json`); that archive is byte-identical to the one that arrived, and manuscript 24 also contains all 176 of its files unchanged. Those copies are not shipped a second time.

Why manuscript 21 is the base, and the order of the Parts. The two manuscripts do not prove the same theorems, so the rule “weaker hypotheses make the base” does not apply. Manuscript 24 depends on manuscript 21 for the count $C_n = 1 + a_n$ and the graph model, cites it as its companion, and answers its first further question; manuscript 21 is therefore Part I and manuscript 24 Part II.

Where the merge had to choose.

- Both articles are printed in full. Part II’s recall of the count and graph model and its re-derivations of (11), (13) and (12) are kept, because Part II’s notation starts there; each is marked by a [write] note (Section 1).
- Manuscript 24’s title block was replaced by the heading of Part II and a [write] paragraph; its abstract and keywords are printed at the head of Part II. Manuscript 21’s title block is the report’s, with a second author line, a merged subtitle and date, and new PDF metadata.
- The two evidence appendices are Appendices A and B; their headings gained “(Part I)” and “(Part II)”.
- The bibliographies are merged. Manuscript 24’s entries for A196460 and for Svatoš et al. are the same works as manuscript 21’s and are printed once, in manuscript 21’s wording

(manuscript 24's said "Consulted" and omitted the PDF address). Manuscript 24's keys `source` and `audit` became `us-source` and `us-audit`. Added: [12, 13, 14, 15]; manuscript 24's [8, 9] are now also cited from Part I.

- No symbol was renamed (Section 2).

What the write changed in the delivered texts.

- Labels: manuscript 21's 65 labels gained the prefix `cta:`, manuscript 24's 61 the prefix `cta:us:`, with every reference updated; this also separates the three names both used (`eq:alpha`, `eq:sectors`, `sec:evidence`). The write added nine labels: `cta:sec:intro` on Part I's unlabelled first section, the front-matter labels `cta:sec:guide`, `cta:sec:notation`, `cta:tab:notation`, `cta:sec:relation`, `cta:sec:limits`, the Part labels `cta:part:one`, `cta:us:part`, and `cta:app:provenance`; 135 in all.
- Added: the editorial note, the table of contents, Sections 1–4, the two Part headings with their [write] paragraphs, [write] notes in Sections 5, 6, 9, 10, 11, 12, 13, 14, 15, 18, 19, 20 and 22, in both evidence appendices and in seven bibliography entries, this appendix, and four bibliography entries.
- No delivered sentence was changed or removed. Every non-claim of both manuscripts stands in place and is collected in Section 4.

Attributions supplied by the write. Manuscript 21 restates Report 69's classification and names it only as accepted source material [3, 4]; the write names the Report and where it is printed [12]. Manuscript 21 applies, without citing them, the connected-graph method of [13, Chapter 8] and the exact-threshold rule of [14, Theorem J.43]; the write credits both (Section 3). Manuscript 21 does not name OEIS A047863 and A002027 for its b_k and c_k ; the write does, as manuscript 24 already did.

Stale or open statements. Part I's question "Growing truncation order" is marked as answered by Part II for the exact sector tail and open for the logarithmic and inverse expansions. No sentence of either manuscript says that the repository lacks something; manuscript 21's finding that the fetched ProveIt pages contained no A196460 was still true at the write (only this report and the Hilbert's-tenth intake notes mention A196460).

Rights. The sequence values printed here (A196460, A047863, A002027) are computed by the shipped programs and agree with the OEIS, whose data are available under the Creative Commons Attribution-ShareAlike 4.0 licence. No OEIS submission is drafted, and nothing was submitted.

Files. The shipped code, data and audit files, the delivery names they still use, the files that were not shipped and how to rerun the checks are listed in the report's `README.md`.

References

- [1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A196460. Entry by Paul D. Hanna, 2 October 2011; leading asymptotic credited to Václav Kotesovec, 25 June 2013. <https://oeis.org/A196460>. Accessed 4 October 2026.
- [2] M. Svatoš, P. Jung, J. Tóth, Y. Wang and O. Kuželka, *On Discovering Interesting Combinatorial Integer Sequences*, arXiv:2302.04606v1, 2023. Appendix A.1, Table 3, printed p. 32. <https://arxiv.org/abs/2302.04606v1>; <https://arxiv.org/pdf/2302.04606v1>.
- [3] *A one-positive-witness alternative, and exact zero-witness classification*, accepted mathematical source note, 4 October 2026, Sections 1–2 and 6. Retained as `ONE_WITNESS.md` in the pinned closure dependencies. SHA-256: 9135dd0e829adb3190e99ed1b62f8d419f9f7540cfd92a1acd1635726fa7172c. [write] Report 69’s note; shipped in [12] as `38-low-arity-la-ONE_WITNESS.md`.
- [4] *Independent low-arity audit*, accepted audit dossier, 4 October 2026, especially Section 8. Retained as a pinned text dependency. SHA-256: 6bf14006170df1e126fb98d8e06ee97cfb51786f9ce11bcc0a37370a896d5afe. [write] Report 69’s audit; shipped in [12] as `38-low-arity-audit-la-AUDIT.md`.
- [5] *Zero-auxiliary clipping tables and A196460*, mathematical proof and exact-evidence packet, 4 October 2026. Complete proof, source notes, checks and preservation ledger retained in `inputs/asymptotics`. Manifest and proof hashes appear in Appendix A. [write] Shipped here as `21-fixed-src-*`.
- [6] *Independent audit of clipping table counts and asymptotic inverses*, 4 October 2026. Complete audit and exact rational evidence retained in `inputs/independent-audit`. Manifest and audit hashes appear in Appendix A. Numerical ratios in Table 2 are transcribed from its `evidence/enclosures.json`. [write] Shipped here as `21-fixed-audit-*`; the enclosures as `data/21-fixed-audit-ev-enclosures.json`.
- [7] *Zero auxiliary clipping tables: Exact counts and all orders asymptotics of A196460*, companion article, 4 October 2026. Previously delivered fixed-order article; retained as an unchanged reference in the source-evidence dossier. Sections 2–4 give the classification, count, and graph interpretation; later sections treat fixed-order logarithmic and inverse expansions. [write] Printed in this report as Part I; its Sections 2–4 are Sections 6–8 here.
- [8] OEIS Foundation Inc., A047863, labeled graphs with two named proper colors. The entry records the binomial-sum formula and parity-dependent theta asymptotic, the latter credited to Václav Kotesovec, 24 June 2013; it also links the classical colored-graph literature. <https://oeis.org/A047863>. Consulted 4 October 2026.
- [9] OEIS Foundation Inc., A002027, connected labeled graphs with two named proper colors. The value at zero uses the entry’s convention. <https://oeis.org/A002027>. Consulted 4 October 2026.
- [10] *Uniform growing-order sector truncations for A196460*, accepted proof and exact-evidence packet, 4 October 2026. Full packet retained in `inputs/uniform-sectors`; proof and manifest pins appear in Appendix B. [write] Shipped here as `24-unif-src-*`.

- [11] *Independent audit of uniform growing-sector truncations*, 4 October 2026. Full analytic audit, fresh exact evidence, and preservation records retained in `inputs/independent-audit`; audit and manifest pins appear in Appendix B. [write] Shipped here as `24-unif-audit-*`.
- [12] [write] *Collision Geometry Is Linear: Diophantine Certificates for Rational Signal Machines and Conservative Particle Dynamics*, research report in ProveIt, `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/signal-machine-collision-certificates/`, Part XIII, source 38: Report 69, *Low-arity finite-clipping Diophantine compilers* (batch-91 manuscript 20, placed by commit `d750d98dd`).
- [13] [write] *Combinatorial Transseries and Their Inverses*, consolidated companion volume in ProveIt, `Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/`, Chapter 8 (`ct:ch:graphs`), Theorem 8.1 (`t2:thm:graphs`), Corollary 8.2 (`t2:cor:graph-range`), Theorem 2.12 (`t1:thm:certificate`).
- [14] [write] V. Reshetnikov, *Transseries: The polynomial-logarithmic calculus, series reversal at infinity, and the inversion of rapidly growing functions*, ProveIt canonical volume, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`, Appendix J, Section J.7, Theorem J.43 (`p0:thm:staircase`).
- [15] [write] *Scoped arithmetic/compiler intake at 0d7f51c44*, review note of ProveIt’s Hilbert’s-tenth research programme, `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_new_arithmetic_0d7f51c44.md`, commit `a9ab9a698`, 4 October 2026.